Asymmetric Labor Income Risk and Stock Market Participation: Evidence from the US

Tai Lo Yeung
Università della Svizzera italiana

Job Market Paper · October 2026

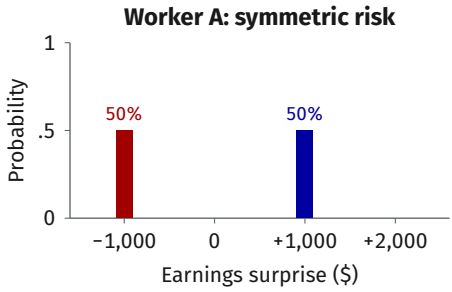
Can earnings opportunities help people start investing?

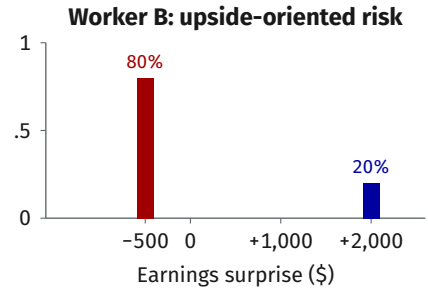
- **Why entry matters.** Starting to own stocks gives households another way to build wealth.
- **The familiar concern.** Uncertain pay can make taking investment risk harder.
- **Variance remains a common summary.** It measures the size of earnings fluctuations, treating gains and losses symmetrically.

  Recent examples: d'Astous & Shore (JFE, 2024); Bacher (RFS, 2025).

Research question: Does the balance of upside and downside earnings variation predict who becomes an investor?

Same variance. Same decision to invest?





Same mean surprise: \$0 Same variance: $\$1{,}000^2$

B has smaller frequent losses and larger rare gains.

More upside predicts stock market participation

U.S. Survey of Income and Program Participation (SIPP)
27,419 annual transitions: 2019–2023 baselines, 2020–2024 outcomes.

- **Finding.** At similar mean growth, total variance and measured resources, more upside predicts entry into stocks or mutual funds outside retirement accounts.
- **Economic size.** A one-standard-deviation increase in earnings upside balance across worker groups predicts **1.60 percentage points more entry**.

  About **16 more entrants per 1,000** initial nonholders.
  Annual entry rate in this sample: 8.82%.
- **Model mechanism.** Realized earnings can finance entry. The anticipated earnings distribution changes saving and resources available for later entry.

▶ Economic contribution ▶ Coefficient units

Literature: what does earnings asymmetry add?

- **The amount of risk remains central.**
  d'Astous & Shore (JFE, 2024); Bacher (RFS, 2025)
  Earnings variance predicts participation and risky shares. Life-cycle models use persistent and temporary income-shock variances.
- **The shape of risk also matters.**
  Catherine, Sodini & Zhang (JF, 2024); Shen (Management Science, 2024)
  Downside earnings risk in poor stock markets matters for stockholding and saving.
- **Earnings risk over a career.**
  Gálvez & Paz-Pardo (JFE, 2026)
  Richer earnings dynamics change participation, risky shares and estimated risk aversion.

This paper: the balance of upside and downside predicts market entry, at similar mean growth, total variance and resources. The model tests that relationship without targeting it.

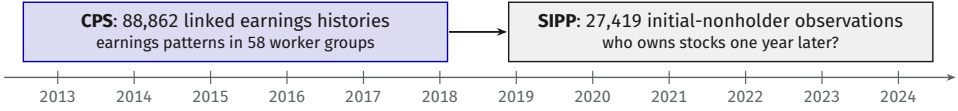
► Earlier evidence ► Closest studies ► Related work in progress

Two public U.S. surveys connect earnings risk to entry

- **Current Population Survey (CPS):** earlier annual wage changes, including workers whose wages fall to zero the next year.
- **Survey of Income and Program Participation (SIPP):** later household assets and stock-market entry.

58 supported earnings cells: industry × education × age band.
Education: high school or less / some college / college. Age: 25–44 / 45–65.

Industry captures the work environment; education and age distinguish skills and career stage. Groups balance these differences against available sample size.



Entry: no stocks or mutual funds today, ownership next year, outside retirement accounts.

▶ How earnings are measured ▶ Samples ▶ Timing

Separate upper and lower semivariance

Start with age-adjusted wage growth ϵ in cell c, with mean μ_c.

$$V_c^+ = \mathbb{E}_c[(\epsilon - \mu_c)^2 \mathbf{1}\{\epsilon > \mu_c\}], \qquad V_c^- = \mathbb{E}_c[(\epsilon - \mu_c)^2 \mathbf{1}\{\epsilon < \mu_c\}].$$

Total variance

$$V_c = V_c^+ + V_c^-$$

Upside balance

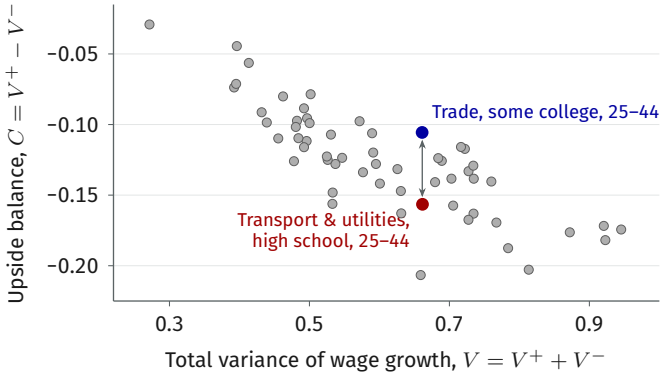
$$C_c = V_c^+ - V_c^-$$

The regression holds total variance fixed. Higher C then means more upper semivariance and less lower semivariance.

Without fixing V, C can rise through either component. It describes the balance of variation, not the chance of an earnings gain.

▶ Construction and centering ▶ Numerical example ▶ Relation to skewness

Why do high-variance cells have more downside?



58 worker groups

All have $C < 0$: more upside means less downside dominance.

$\mathrm{corr}(C, V) = -0.81$.

Cells with more transitions to zero annual wages have higher V and lower C.

Highlighted pair: almost equal variance, different balances of gains and losses.

▶ Economic drivers ▶ Highlighted pair ▶ Identify every cell

Who enters among people who own no stocks today?

Sample: initial nonholders only.

$$\underbrace{1\{\text{Participates}_{i,t+1}\}}_{\text{entry next year}} = \alpha + \beta \frac{C_{c(i,t)} - \bar{C}}{s_C} + \text{controls}.$$

What is held comparable?	Variables in the regression
Total earnings variance	V and $\sqrt{V}$, allowing a curved relationship
Mean earnings growth	Mean growth and its square
Current resources	Initial assets, earnings and nonpositive-earnings indicator
Demographics	Age, age squared and sex
Group and year effects	Industry, education, age band and outcome year

Participation means stocks or mutual funds outside retirement accounts. Weighted linear probability model; standard errors clustered by industry–education group.

▶ Full specification ▶ Measurement versus controls

Upside balance and stock market participation

Initially nonparticipating households	Estimate
Entry difference per source SD	$+1.60$ pp
Clustered standard error	0.51 pp
Mean annual entry	8.82%
Initial nonholder transitions	27,419

About 16 more entrants per 1,000 initial nonholders.

Roughly 18% of the mean entry rate, at similar variance and resources.

▶ All earnings measures ▶ Coefficient units ▶ Published magnitudes

Does upside balance matter with a simpler variance control?

Earnings terms	V	$V, \sqrt{V}$	C, V	$C, V, \sqrt{V}$
Upside balance C	—	—	1.111^{**}	1.601^{***}
			(0.526)	(0.506)
Variance V	-0.457	-11.839^{***}	0.053	-15.380^{***}
	(0.584)	(3.845)	(0.548)	(3.505)
Standard deviation $\sqrt{V}$	—	11.457^{***}	—	15.762^{***}
		(3.689)		(3.456)

The positive association survives the linear specification.

C, V is equivalent to entering V^+, V^- directly. Adding $\sqrt{V}$ allows a curved variance relationship, with one additional coefficient.

Same 27,419 transitions, weights and other controls. Coefficients in pp per own source SD; CR1/t_{29} SEs. $^{**}p < 0.05$, $^{***}p < 0.01$.

▶ Direct upper/lower coefficients ▶ Why opposite variance signs?

What do the additional comparisons establish?

Question	Evidence and interpretation
Measurement and sorting	Positive entry estimates under alternative earnings measures, richer group controls and reporting screens.
Another survey and period?	Panel Study of Income Dynamics (PSID): no clear association over ages 25–65; a positive estimate at 25–44 comes from an exploratory split.
Only short-lived entry?	SIPP: the association also appears for entry followed by later ownership.
Do stated beliefs line up?	New York Fed SCE: directly reported earnings expectations and changes in retirement portfolios.

The surveys answer different questions: entry, external comparison, and subjective expectations.

▶ Controls and reporting ▶ PSID ▶ Samples ▶ Earnings measures

Entry followed by later ownership

Within each column: the same nonholders, two outcomes.

Follow-up cohort	One later year	Two later years
Entry alone (pp)	2.09^{***} (0.70)	3.31^{**} (1.43)
Entry + later ownership (pp)	0.98^{*} (0.50)	1.89^{**} (0.79)
Sustained outcome $= 1$	$0, 1, 1$	$0, 1, 1, 1$
Initial nonholder transitions	9,236	3,254

The sample includes all initial nonholders with the required annual follow-up.

Other observed sequences have sustained outcome 0.

▶ Entry and interruption ▶ Sample and sequences

Do people's own earnings expectations line up with investing?

New York Fed Survey of Consumer Expectations (SCE). Directly reported beliefs complement the historical worker-group measures.

Belief measure	Reallocation	Equity share
Upside-minus-downside tilt	0.028^{**}	-0.576
	(0.013)	(0.664)
Sum of semideviations	-0.017	-3.805^{***}
	(0.018)	(1.011)
Respondents	2,136	3,332

More upside is associated with less movement toward safer retirement assets. Its association with the standing equity share is imprecise.

Reallocation: $+1$ riskier, -1 safer, 0 unchanged. Equity share: percentage points. Controls: expected growth, age, education, income, state and module year; 50 state clusters. Belief and action windows overlap. This is portfolio adjustment, not market entry.

▶ Different outcomes and samples ▶ Inference

Can a standard saving model explain the entry pattern?

- **A familiar mechanism.** Earnings add to cash. With a participation fee, investing in stocks becomes worthwhile when enough can be saved.
- **A role for the earnings distribution.** Equal means and variances can conceal different chances of reaching that point.
- **This paper's test.** Estimate the model from earnings and financial outcomes, then ask whether it generates the observed entry association.

The mechanism is standard. The 58 cells provide measured earnings differences.

The entry–risk coefficients are excluded from estimation.

▶ Related models ▶ Estimation sequence ▶ Alternative access model

Two dates: when is paying the fee worthwhile?

1. **Today:** income Y arrives. Add inherited resources w to get cash $m = w + Y$.
2. **Today:** choose consumption and saving. Pay a fixed fee κ to invest part of saving in stocks.
3. **Tomorrow:** consume the return on saving. Stocks offer higher expected returns, with risk.

Optimal entry occurs above a cash threshold m^*.

$$\Pr(\text{entry}) = \Pr\left(Y \geq \underbrace{m^* - w}_{\text{income needed to reach the threshold}}\right).$$

The fixed fee absorbs a smaller share of resources when cash is higher.

▶ Equal-variance example ▶ Saving with future earnings ▶ Derivation

Cash today and earnings prospects enter different parts of the choice

Realized resources	Earnings and asset returns have arrived. Cash, earnings state, investment access and the fee offer are known.
Future earnings	The household knows the distribution, and weighs consumption it could afford in each future outcome.
Current choices	Consume, save, and decide whether to hold stocks and what share to invest.

$$c_t + A_t + D_t K_t = M_t, \qquad M_{t+1} = A_t R_{t+1}(\alpha_t) + Y_{t+1}.$$

M: cash; c: consumption; A: saving; D: stockholding indicator; K: fee.

α: stock share; R: gross portfolio return; Y: disposable income.

Expectations change today's saving; saving changes tomorrow's resources.

▶ Resources, expectations and saving ▶ Numerical example ▶ Optimization

What determines a worker's earnings?

While employed, log wages combine a career profile and two shocks.

$$\log W_{gt} = \underbrace{f_g(t)}_{\text{predictable profile}} + \underbrace{\eta_t}_{\text{persistent state}} + \underbrace{\varepsilon_{gt}}_{\text{temporary shock}} .$$

Career profile	Typical pay at that age, education and industry
Persistent state	Small ordinary innovations plus occasional promotions or earnings losses that carry forward
Temporary shock	Variation in current pay that does not carry forward

A separate employment transition determines whether wages are zero.

Disposable income also reflects taxes, own retirement contributions and resources outside wage work.

▶ Where asymmetry enters ▶ Exact earnings equation

How do 58 measured cells enter the model?

An earnings process gives probabilities of future pay and wage work.

Shared foundation	College and noncollege career processes
30 career groups	Industry $\times$ education, with group wage levels
58 measured cells	Adjustments before and after age 45 within those groups

Each cell changes four features: **mean growth**, **leaving wage work**, **temporary variation**, and the **balance of favorable and adverse shocks**.

Preferences, participation costs and opportunity probability are common.

Two unsupported group–age combinations use the other age band's adjustment.

▶ Four fitted adjustments ▶ How tails differ ▶ Why 58, not 580?

Structural estimates and external inputs

Inputs held fixed

Input	Value / source
Safe gross real return	1.02
Ordinary stock return: mean / SD	1.06 / 0.18
Annual crash probability	1%
Crash net real return	-48.5%
EIS ψ	0.5
Earnings / resources	CPS / PSID
Observed initial states	SIPP

Estimated financial parameters

Parameter	Estimate
Risk aversion γ	8.801
Discount factor β	0.811
Mean fee offer κ_R	**$524**
Log-cost standard deviation σ_κ	0.944
Opportunity probability q	**0.176**

Simulated method of moments: five parameters, eight financial targets.

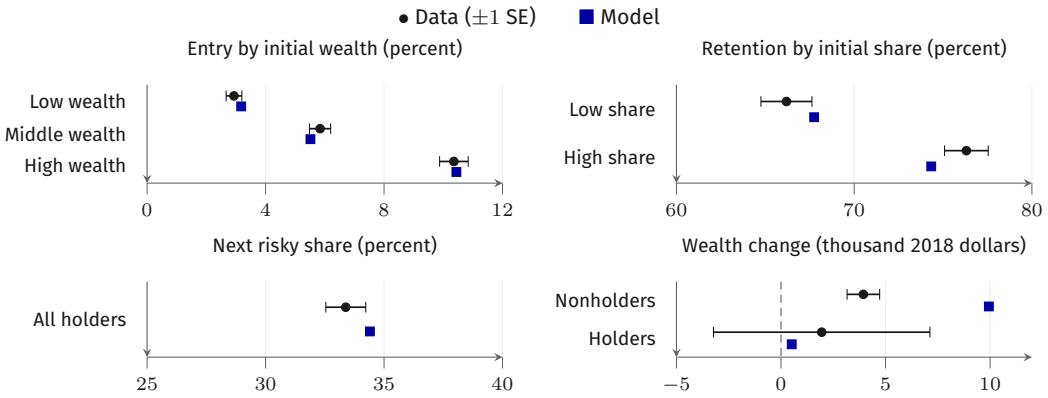
Calibrated: returns and EIS. **Estimated separately:** earnings processes.

From data: resource profiles and initial states. **Untargeted:** entry–risk coefficients.

▶ Targets and weighting ▶ Financial fit ▶ Opportunities and fees

How well does the model fit its financial targets?

Eight financial moments; five estimated parameters



Main discrepancy: excessive wealth growth among initial nonholders.

Next: test the entry–risk relationship, which was excluded from fitting.

◀ Back ▶ Full target table ▶ Targets and weighting

What changes when we pool earnings processes?

	Cell processes	Pooled processes
Earnings adjustments	Group and age specific	Removed
Career foundation	Same education processes and group wage levels	
Starting point	Same observed households and initial resources	
Preferences and costs	Same estimated financial parameters	
Decisions	Solve optimal choices under each earnings process	

What do measured earnings differences add, with financial parameters fixed?

Pooling changes the full earnings distribution, including mean growth. It does not isolate asymmetry, and the pooled model is not re-estimated.

▶ Untargeted comparison ▶ Same regression ▶ What remains under pooling

Measured earnings differences strengthen the entry association

Upside-balance coefficient: percentage points per source SD

Same 17,606 initial nonholders	Entry association
Observed data	1.816^{***} (SE 0.480)
Model with pooled earnings processes	-0.005
Model with cell earnings processes	$+0.241$

Cell processes add **0.247 pp** with financial parameters fixed.

The model reproduces the positive sign, with a smaller magnitude.

Same entry regression, controls and weights. These coefficients were not targeted.

▶ Resources and decision rules ▶ Matched comparison ▶ Full comparison

Why does the cell model produce a stronger entry association?

Change realized earnings: use the other earnings process to generate next pay and work states; keep a previously solved decision rule.

Change the decision rule: keep realized earnings the same; use choices optimized for the other future earnings process.

Contribution to the upside-balance coefficient	pp per source SD
Earnings outcomes, resources and states	$+0.274$
Different optimal decision rules	-0.027
Total: cell minus pooled	$+0.247$

Realized earnings differences dominate this one-year comparison.

The next illustration shows why saving responses matter over several years.

◀ Back ▶ Four comparisons ▶ What is a decision rule?

An illustration: shift variance toward upside or downside

A separate experiment within the fitted cell earnings processes.

Let U_y and L_y be the upper and lower partial variances of earnings in dollars.

More upside: $(U_y + d, L_y - d)$ More downside: $(U_y - d, L_y + d)$.

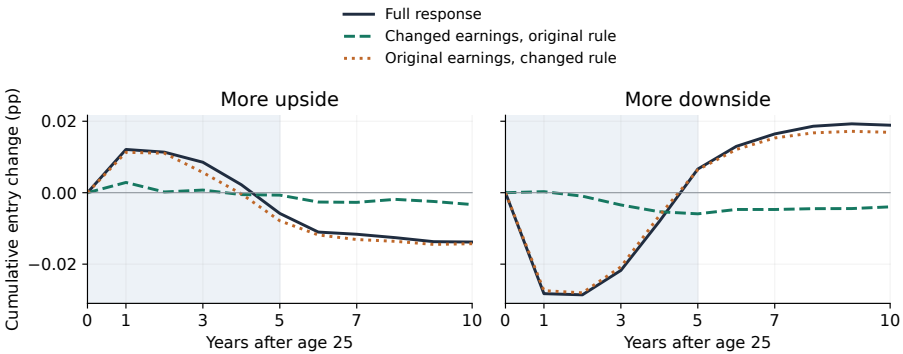
- d transfers variance, not income. Expected disposable earnings and total dollar variance stay fixed.
- **Years 1–5 (ages 26–30):** draw from the changed distribution each year. **Year 6 onward:** baseline. Households know this schedule.
- Parameters, starting states and opportunity/return draws fixed; risk coefficients untargeted.

Follow 395 initial nonholders from age 25 for ten years. The shock illustrates the mechanism; its size is not fitted to the empirical coefficient.

Dollar moments are fixed; growth-rate moments can change.

▶ Entry ▶ Saving at the same cash

Illustration: more upside can raise entry first, then lower it



The changed saving and investment plan drives most of the later reversal.
Years 1–5: changed earnings distribution. Year 6 onward: baseline. Households know this schedule.

▶ Wealth ▶ Saving at the same cash

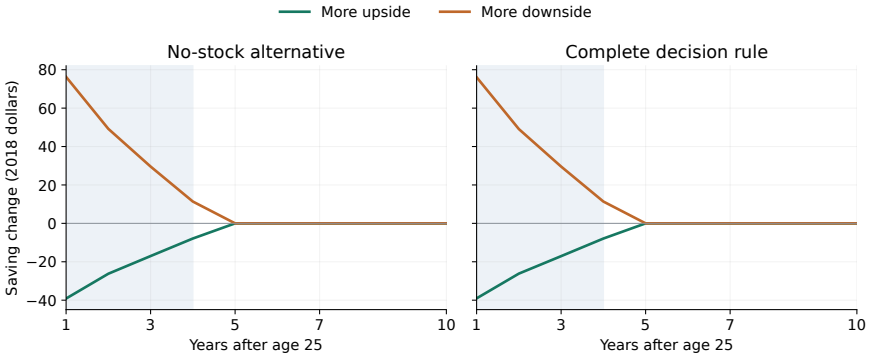
In this illustration, upside reduces financial wealth



The effect persists through the financial states households carry forward.
From age 30, rules coincide at identical cash and earnings states; past choices still affect wealth.

▶ Same-cash comparison ▶ Consumption and fees

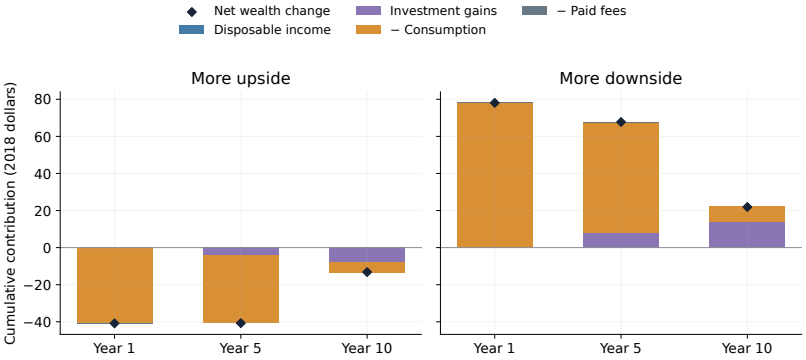
In this illustration, upside lowers saving at the same cash



First year: saving falls by about $39, even in the no-stock alternative.
Future earnings risk changes desired buffer saving, even with expected earnings fixed.

▶ Where the money goes ◀ Back

Higher consumption accounts for the initial wealth decline



First-year upside response: consumption rises $40.81; fees rise $0.03.
Expected earnings stay fixed. Households spend more and leave less saving to finance later entry.

◀ Back ▶ One year or lasting prospects? ▶ Model assessment

Earnings opportunities help explain who becomes an investor

- **Finding.** More upside predicts entry beyond mean growth, total variance and measured resources.
- **Economic size.** 1.60 percentage points per source SD of upside balance, about 16 more entrants per 1,000 initial nonholders.
- **Distinct question.** Earnings asymmetry itself, measured without stock returns, predicts subsequent entry and entry followed by ownership.
- **Model lesson.** Realized earnings can fund entry now. Future earnings risk changes saving and the resources available for later entry.

The entry relationship provides an untargeted test beyond the model's financial fit.

▶ Closest empirical papers ▶ Relation to skewness ▶ Discussion appendix

Discussion questions

Empirical evidence

- ▶ How are the measures constructed?
- ▶ What drives the differences across cells?
- ▶ Which cells are in the scatter?
- ▶ Why not regress on the two tails?
- ▶ What are the controls?
- ▶ Alternative earnings measures
- ▶ PSID evidence
- ▶ Which ownership sequences count?

Model mechanisms

- ▶ Why 58 cells?
- ▶ Earnings process equations
- ▶ The household optimization problem
- ▶ Two-cell numerical example
- ▶ Resources versus decision rules
- ▶ One year, five years or working life?
- ▶ All duration decompositions
- ▶ Financial fit and discrepancies

◀ Back

Why do earnings distributions differ across cells?

Across 58 cells: correlation with zero-wage share	Correlation
Total variance V	$+0.92$
Upside balance C	-0.85

Transitions to zero annual wages help explain the scatter's orientation.

- The wage measure includes transitions from positive annual wages to zero. Zero wages need not mean unemployment or dismissal.
- Annual earnings also reflect hours, job changes and wage changes. These records do not separate promotions from persistent losses.
- The source period precedes the participation outcomes. This cross-cell comparison does not identify a business-cycle shock.

◀ Back ▶ Does zero-wage risk absorb the result? ▶ Persistence and cycle evidence

Does the zero-wage share absorb the participation association?

Same 27,419 initial nonholders, controls, weights and 30 industry–education clusters.

Coefficient, pp per source SD	Baseline	Add zero-wage share
Upside balance C	1.601^{***}	2.420^{**}
	(0.506)	(0.911)
Zero-wage share	—	2.630
		(2.030)

The positive association remains after controlling for the frequency of zero wages.

The larger standard error reflects a demanding comparison among strongly related cell measures. The coefficient increase does not identify a causal channel.

Outcome: next-year stock market participation among initial nonholders. Clustered SEs in parentheses, t_{29} inference. $^{**}p < 0.05$, $^{***}p < 0.01$.

◀ Back ▶ Two cells at similar variance

The highlighted cells: similar variance, different downside exposure

Both age 25–44. Semivariances use age-adjusted annual arc wage growth.

Education	Trade Some college	Transport / utilities High school
Upper semivariance V^+	0.278	0.253
Lower semivariance V^-	0.383	0.409
Total variance V	0.661	0.661
Upside balance C	-0.106	-0.156
Zero annual wages next year	5.67%	7.75%
Linked earnings histories	2,493	802

Trade has less lower-tail and more upper-tail variation.

The balance gap is about 1.5 source SD. These descriptions do not identify the job events behind the differences.

◀ Back ▶ All 58 cells

Does measured cyclical exposure absorb the association?

Saved checks: add each control to the same participation regression.

Additional control	Upside balance, pp	SE
None	1.601^{***}	0.506
Cell's mean wage growth in 2020	1.783^{***}	0.528
Mean-growth exposure to unemployment changes	1.612^{***}	0.479
Lower-semivariance exposure	1.701^{***}	0.533
Upside-balance exposure	1.662^{***}	0.487
All three unemployment exposures	2.104^{***}	0.452

The positive association remains with these measured cycle controls.

PSID exposures: slopes of biennial cell moments on national unemployment changes, 1999–2019. Same 27,419 SIPP transitions, 30 clusters, CR1/t_{29}. $^{***}p < 0.01$.

◀ Back ▶ Permanent or temporary changes?

Can we distinguish permanent losses from temporary changes?

- **Annual CPS histories do not label the underlying event.** Wage changes can combine job moves, hours changes and changes in wage rates.
- **An exploratory transitory-share comparison is sensitive to measurement.** Its positive interaction weakens when extreme earnings changes are trimmed.
- **Longer PSID histories discipline persistence in the model.** They do not establish which event caused a particular CPS cell's position in the scatter.

The evidence supports a measured earnings-distribution comparison.

It does not assign causal shares to promotions, permanent losses and temporary earnings changes.

◀ Back ▶ Model earnings components

Which cells are in the scatter? (1/6)

$V = V^+ + V^-$; $C = V^+ - V^-$. HS: high school or less. Zero: no wages next year.

Industry	Education	Age	V^+	V^-	Zero %	Histories
Agriculture	HS	25-44	0.305	0.508	12.2	623
Agriculture	HS	45-65	0.348	0.524	17.7	565
Construction	College	25-44	0.247	0.379	6.2	491
Construction	College	45-65	0.303	0.432	8.8	436
Construction	HS	25-44	0.299	0.468	10.4	1,950
Construction	HS	45-65	0.286	0.449	10.6	1,661
Construction	Some col.	25-44	0.282	0.421	7.7	867
Construction	Some col.	45-65	0.274	0.431	10.0	767
Manufacturing	College	25-44	0.162	0.233	2.9	1,520
Manufacturing	College	45-65	0.192	0.290	4.4	1,623

CPS origins 2013–2017. Weighted moments; unweighted history counts.

◀ Back ▶ Next cells

Which cells are in the scatter? (2/6)

$V = V^+ + V^-$; $C = V^+ - V^-$. HS: high school or less. Zero: no wages next year.

Industry	Education	Age	V^+	V^-	Zero %	Histories
Manufacturing	HS	25-44	0.201	0.326	5.1	1,936
Manufacturing	HS	45-65	0.201	0.324	6.2	2,694
Manufacturing	Some col.	25-44	0.201	0.296	3.9	1,228
Manufacturing	Some col.	45-65	0.170	0.269	4.3	1,478
Trade	College	25-44	0.234	0.361	5.4	1,822
Trade	College	45-65	0.302	0.420	7.3	1,627
Trade	HS	25-44	0.282	0.407	7.2	3,490
Trade	HS	45-65	0.269	0.410	8.1	3,328
Trade	Some col.	25-44	0.278	0.383	5.7	2,493
Trade	Some col.	45-65	0.242	0.348	6.0	1,855

CPS origins 2013–2017. Weighted moments; unweighted history counts.

◀ Back ▶ Next cells

Which cells are in the scatter? (3/6)

$V = V^+ + V^-$; $C = V^+ - V^-$. HS: high school or less. Zero: no wages next year.

Industry	Education	Age	V^+	V^-	Zero %	Histories
Transport/utilities	College	25-44	0.202	0.290	3.6	365
Transport/utilities	College	45-65	0.221	0.355	4.5	445
Transport/utilities	HS	25-44	0.253	0.409	7.7	802
Transport/utilities	HS	45-65	0.234	0.397	7.8	1,144
Transport/utilities	Some col.	25-44	0.280	0.447	11.5	635
Transport/utilities	Some col.	45-65	0.188	0.345	7.5	784
Information	College	25-44	0.170	0.262	3.2	473
Information	College	45-65	0.212	0.290	4.0	428
Information	HS	25-44	0.187	0.297	5.3	360
Information	HS	45-65	0.188	0.304	4.1	411

CPS origins 2013–2017. Weighted moments; unweighted history counts.

◀ Back ▶ Next cells

Which cells are in the scatter? (4/6)

$V = V^+ + V^-$; $C = V^+ - V^-$. HS: high school or less. Zero: no wages next year.

Industry	Education	Age	V^+	V^-	Zero %	Histories
Information	Some col.	25-44	0.176	0.220	2.1	357
Information	Some col.	45-65	0.212	0.319	5.9	368
Finance/real estate	College	25-44	0.189	0.291	3.7	1,799
Finance/real estate	College	45-65	0.300	0.416	6.6	1,546
Finance/real estate	HS	25-44	0.226	0.433	10.1	531
Finance/real estate	HS	45-65	0.236	0.355	6.8	683
Finance/real estate	Some col.	25-44	0.192	0.341	6.0	842
Finance/real estate	Some col.	45-65	0.229	0.371	7.5	923
Professional/business	College	25-44	0.201	0.300	4.2	3,676
Professional/business	College	45-65	0.280	0.404	7.6	2,986

CPS origins 2013–2017. Weighted moments; unweighted history counts.

◀ Back ▶ Next cells

Which cells are in the scatter? (5/6)

$V = V^+ + V^-$; $C = V^+ - V^-$. HS: high school or less. Zero: no wages next year.

Industry	Education	Age	V^+	V^-	Zero %	Histories
Professional/business	HS	25-44	0.298	0.486	10.6	1,602
Professional/business	HS	45-65	0.310	0.450	10.2	1,774
Professional/business	Some col.	25-44	0.298	0.437	7.4	1,663
Professional/business	Some col.	45-65	0.242	0.389	8.1	1,683
Education/health	College	25-44	0.191	0.271	3.5	6,966
Education/health	College	45-65	0.192	0.304	4.8	6,270
Education/health	HS	25-44	0.297	0.430	8.1	1,581
Education/health	HS	45-65	0.212	0.335	6.9	2,458
Education/health	Some col.	25-44	0.237	0.335	5.1	2,989
Education/health	Some col.	45-65	0.205	0.333	5.8	3,038

CPS origins 2013–2017. Weighted moments; unweighted history counts.

◀ Back ▶ Next cells

Which cells are in the scatter? (6/6)

$V = V^+ + V^-$; $C = V^+ - V^-$. HS: high school or less. Zero: no wages next year.

Industry	Education	Age	V^+	V^-	Zero %	Histories
Leisure/hospitality	HS	25-44	0.374	0.546	16.0	616
Leisure/hospitality	HS	45-65	0.386	0.560	14.8	688
Leisure/hospitality	Some col.	25-44	0.370	0.552	13.7	389
Public administration	College	25-44	0.121	0.150	1.0	1,339
Public administration	College	45-65	0.159	0.233	2.9	1,427
Public administration	HS	45-65	0.176	0.302	5.5	653
Public administration	Some col.	25-44	0.178	0.235	3.0	769
Public administration	Some col.	45-65	0.173	0.283	5.0	945

CPS origins 2013–2017. Weighted moments; unweighted history counts.

◀ Back ▶ First cells

A direct regression on upper and lower semivariances

Among initial nonparticipants, regress next-year participation on

$$\alpha + b_+ V_c^+ + b_- V_c^- + \text{mean growth, resources, demographics and fixed effects.}$$

This linear specification omits $\sqrt{V}$. All other controls and the sample are unchanged.

Direct coefficient	Estimate	Clustered SE
Upper semivariance V^+	$+0.331^*$	(0.167)
Lower semivariance V^-	-0.324^{**}	(0.154)

Units: percentage points of participation per 0.01 growth-squared unit of the indicated semivariance, holding the other fixed.

More upper variation predicts more participation; more lower variation predicts less.

One linear regression, equivalently parameterized by C, V. These are not local derivatives of the nonlinear baseline. $N = 27,419$, 30 clusters, CR1/t_{29}. $^*p < 0.10$, $^{**}p < 0.05$.

◀ Back ▶ Controls and sample

What does earnings upside balance add?

How earnings vary helps explain who becomes an investor.

- **Beyond the amount of risk.** Upside balance predicts entry at similar mean growth, variance and measured resources.
- **Earnings asymmetry itself.** The measure uses earnings outcomes, without stock returns. The question differs from exposure to joint downturns.
- **A test for portfolio models.** Can standard saving and participation choices generate this entry pattern when risk coefficients are untargeted?

▶ Measurement comparisons ▶ Closest studies ▶ Related work in progress

Literature: resources and stock-market entry

- **Resources and ownership transitions.**
  Briggs et al. (JFE, 2021); Bonaparte et al. (Management Science, 2025)
  Lottery windfalls raise entry among nonholders. Income and wealth changes also help explain entry and exit.

- **This paper measures earnings opportunities before entry.**
  Earlier earnings distributions, measured in a separate survey, predict entry and entry followed by later ownership.

- **The model tests the same entry relationship.**
  Entry–risk coefficients are excluded from fitting. The model separates changes in resources along earnings paths from changes in saving and investment decisions.

▶ Closest empirical studies ▶ Model comparison

How will we answer the question?

1. **Measure:** where does earnings variation come from?
2. **Evidence:** who starts investing, and who keeps holding?
3. **Checks:** earnings measurement, sorting and reporting
4. **Model:** how do earnings paths and saving decisions shape entry?

Checks tied to specific concerns

Each check changes the baseline separately. The first block requires positive financial assets and valid risky shares at both dates.

Concern	Specification	Estimate, pp (SE)	Nonholders
	Baseline: valid shares at both dates	1.646^{**} (0.599)	24,235
Group-year conditions	Industry $\times$ year; education $\times$ year	1.663^{**} (0.619)	24,235
Persistent group sorting	Instead: industry $\times$ education effects	1.865^{**} (0.875)	24,235
Reporting and household changes	Reported, self-reported, stable households	2.557^{***} (0.555)	11,229
Realized earnings: a separate common sample of 23,261 nonholders			
	Baseline	1.518^{**} (0.587)	23,261
	Add annual earnings level and change	1.574^{**} (0.586)	23,261

Both broad and nonretirement denominators must be positive; their risky shares must lie in $[0, 1]$.

▶ Sorting ▶ Samples

Evidence from the PSID

Purpose: another survey and period. Panel Study of Income Dynamics (PSID), biennial entry among initial nonholders; baseline years 1999–2017.

Age range	Upside balance (pp)	SE
Full sample: 25–65	-0.18	0.39
Exploratory age split		
25–44	2.42^{**}	1.11
45–65	-0.98	1.01

No clear full-sample association; a positive estimate among younger workers.

PSID uses different households. Some outcomes occur before the CPS earnings data, so this is not the same timing design as SIPP.

▶ Later-period age comparison ▶ Period differences

Earnings upside balance and participation decisions

- **Main result:** more upside-oriented variation predicts more next-year entry at similar variance and resources.
- **Later ownership:** the association also appears for entry followed by ownership at later annual observations.
- **Economic question:** how does the earnings environment relate to a nonholder's subsequent investment decision?

The model tests a familiar resource mechanism against this entry pattern.

▶ Entry and retention

Same variance, different chances of reaching the threshold

Earnings innovation	Symmetric	More upside
Possible loss	$-\$1{,}000$ (50%)	$-\$500$ (80%)
Possible gain	$+\$1{,}000$ (50%)	$+\$2{,}000$ (20%)
Mean / standard deviation	\$0 / \$1,000	\$0 / \$1,000
Entry if \$1,500 short	0%	20%
Entry if \$500 short	50%	20%

The gain matters when it carries a household across the threshold.

The sign depends on where households start. Cells are not needed for this mechanism.

Illustrative dollar innovations around common expected earnings, not estimated responses.

◀ Back ▶ Derivation

Can a workhorse model reproduce the entry association?

1. Estimate earnings processes from CPS and PSID earnings moments.
2. Fit five financial parameters to eight SIPP financial outcomes.
3. Solve the same saving and participation problem under cell and pooled earnings dynamics.
4. Apply the empirical entry regression to those predictions.

Untargeted test: do measured earnings environments generate the observed upside balance–entry relationship?

Earnings-risk coefficients never enter the fit. Related comparison: Gálvez & Paz-Pardo (2026).

▶ Initial states ▶ Published precedents

Four adjustments connect each cell to its earnings data

Cell adjustment	What it changes
Departure risk	Probability of leaving wage work
Growth	The predictable earnings path with age
Shock scale	The size of transitory variation and reporting error
Asymmetry tilt a_c	Relative weights on favorable and adverse persistent and temporary shocks

Fit all four adjustments jointly to four CPS earnings moments:

zero-wage frequency, mean growth, upper partial variance and lower partial variance.

PSID longer-horizon moments discipline the shared career process. Stock-market entry coefficients never enter earnings estimation.

▶ How the tilt works ▶ Exact cell mapping

Gains and losses need not be mirror images

Persistent earnings combine ordinary innovations with occasional jumps.

$$\eta_{t+1} = \rho\eta_t + \zeta_{t+1} + \underbrace{J^+ I^+_{t+1}}_{\text{promotion}} - \underbrace{J^- I^-_{t+1}}_{\text{earnings loss}} - m.$$

ρ: persistence; ζ: small normal innovation; m: centering drift.

J^+, J^-: jump sizes; I^+, I^-: indicators for promotion or loss.

Promotion and loss are mutually exclusive. Their sizes and probabilities can differ.

A positive cell tilt a_c shifts relative weight toward favorable shocks.

$$\text{Promotion weight } \propto e^{a_c}, \qquad \text{Loss weight } \propto e^{-a_c}.$$

Temporary pay shocks follow a mixture distribution. The same tilt shifts their relative weights toward the upper tail.

▶ Exact persistent tilt ▶ Temporary-shock tilt

Realized income changes cash. Expectations change the plan.

Realized resources	Earnings already received add to cash available for consumption, saving and participation fees.
Expectations	The future earnings distribution changes how valuable a financial buffer and stock investment are today.
Saving over time	Today's saving becomes invested wealth. Its returns and next year's income become next year's cash.

A decision rule gives the optimal choice for each current state.

At identical cash and other states, different future risks can change consumption, saving and entry.

Expected pay can stay fixed. The value of a buffer also depends on what happens in bad earnings outcomes.

▶ Continuation values ▶ Earnings and decision rules

Two-period illustration: same mean and variance

Earnings today and next year follow one of these two processes.

All amounts are in thousands of dollars.

	Upside cell (U)	Downside cell (D)
Ordinary outcome	85 with 80% chance	115 with 80% chance
Tail outcome	160 with 20% chance	40 with 20% chance
Mean / standard deviation	100 / 30	100 / 30

U has a rare large gain; D has a rare large loss.

Same log utility and patience; entry costs 1.5. Safe saving earns 0%; stocks return -50% or $+70\%$, each with 50% chance, independently of earnings.

These are illustrative cells, separate from the fitted 58-cell model.

▶ Full assumptions ▶ Same cash, different prospects

Two-period illustration: expectations at the same cash

Give both households 100 in cash today. Only the future distribution differs.

Future earnings process	Consume today	Save safely	Enter stocks?
Upside cell (U)	96.96	3.04	No
Downside cell (D)	92.77	7.23	No

The downside household saves 4.19 more as a buffer.

Both expect 100 tomorrow. D can earn only 40, while U earns at least 85. The buffer is more valuable in D despite equal means and variances.

This isolates the response to future risk. Current cash, preferences, entry fees and investment opportunities are identical.

All amounts are in thousands of dollars. Uncalibrated optimal choices.

◀ Back ▶ Let earnings arrive ▶ Full assumptions

Two-period illustration: let earnings become cash

Start with no inherited assets. Current and future earnings are independent draws from the same cell process. Amounts are in thousands of dollars.

Current outcome	Chance	Cash	Optimal saving	Enter?
U: ordinary	80%	85	0	No
U: large gain	20%	160	32.22 in stocks	Yes
D: large loss	20%	40	0	No
D: ordinary	80%	115	14.02 safely	No

The large gain finances stock entry despite the fee.

U: 20% enter; mean saving is 6.44. D: 0% enter; mean saving is 11.22.

The group saving gap combines realized-resource and expectation differences. It is not an isolated precautionary-saving effect.

◀ Back ▶ Full assumptions

Apply the same regression to model entry probabilities

	Data	Model
Outcome	Entry observed: 0 or 1	Predicted entry probability
Households	The same 17,606 initial nonholders	
Right-hand side	Same risk measures, controls, fixed effects and weights	

Average over earnings, opportunities and costs, conditioning each origin on next-year wage work and positive assets.

Compare gross cell predictions with data.

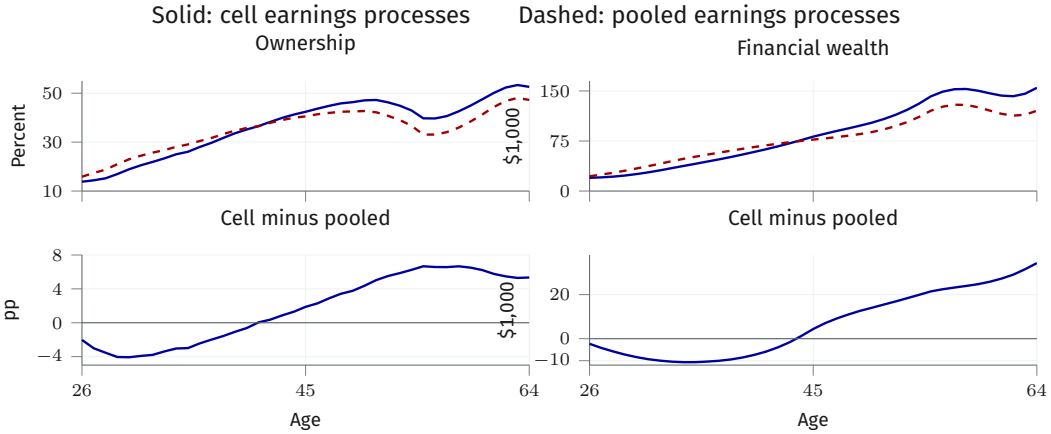
Subtract pooled predictions to describe the contribution of cell earnings processes.

The matched data coefficient is 1.816 pp; the full-sample headline is 1.601 pp.

▶ Why probabilities do not shrink coefficients ▶ Matched sample

Ownership and wealth paths

Same 135 age-25 origins; two college groups; simulated paths



Why five years? Compare three durations

Five years is an illustrative horizon, not an estimated duration.

Duration	Affected ages	What follows
One year	26	Baseline earnings from age 27
Five years	26–30	Baseline earnings from age 31
Remaining working life	26–64	Retirement income unchanged

Same annual shock at shared ages; expected disposable earnings and dollar variance stay fixed.

Same 395 initial nonholders and fixed parameters; risk coefficients remain untargeted.

Households know the schedule. Starting portfolios are given; the first new choice follows year 1 earnings.

◀ Back ▶ Entry comparison ▶ Wealth comparison

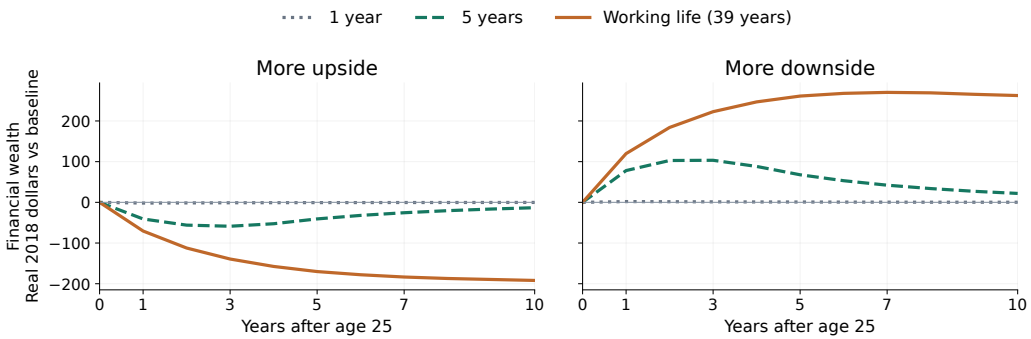
Does the entry response depend on duration?



The five-year reversal depends on how long earnings prospects change.
One-year entry is near zero and grid-sensitive. Cumulative entry under lasting upside fades toward zero. Its later annual first-entry differences are negative.

◀ Back ▶ Wealth ▶ Annual first entry

How does duration change financial accumulation?



Lasting prospects sustain the saving and wealth response.
At year 10, lasting upside lowers wealth by about $192; lasting downside raises it by $262. The five-year gaps are about $13 and $22.

◀ Back ▶ Why wealth and entry differ ▶ All comparisons

Why can wealth fall while cumulative entry barely changes?

Illustrative upside shift: expected earnings stay fixed.

- **At the same cash and state,** changed future risks make entry slightly more attractive and reduce desired buffer saving.
- **Saving then changes future resources.** More consumption leaves less invested wealth and lower investment gains.
- **Cumulative entry retains the early entrants.** Fewer first-time entrants later roughly offset the initial extra entrants.

Five years: the entry encouragement ends, but the smaller buffer remains. **Lasting change:** both forces continue.

◀ Back ▶ Entry at the same cash ▶ Annual entry ▶ All diagnostics

What do the model comparisons teach us?

- **Familiar mechanism.** Earnings can help households reach the resources needed to enter.
- **Measured environments.** Cell dynamics strengthen the untargeted entry association, mainly through realized resources.
- **Controlled illustration.** Future earnings risk changes saving at the same cash, changing resources for later entry.
- **Quantitative comparison.** Upside balance gradient: 0.241 pp with cell processes, against 1.816 pp in the matched data.

Nonholders accumulate too much wealth and former holders re-enter too slowly.

▶ Cell versus pooled ▶ Financial discrepancies ▶ Saving illustration

Earlier evidence on income risk and portfolios

Study	Empirical risk measure
Guiso, Jappelli & Terlizzese (1996, AER)	Subjective income uncertainty
Angerer & Lam (2009, JF)	Permanent and transitory income variability
Fagereng, Guiso & Pistaferri (2018, ReStud)	Firm-transmitted uninsurable wage risk

This paper studies the balance of above- and below-mean earnings variation and subsequent entry.

Different income-risk measures and portfolio outcomes imply different empirical comparisons.

◀ Back ▶ Theory and measurement

Removing age profiles and centering are different steps

Group G: industry $\times$ education. Cell c: group $G \times$ age band.

1. Measure annual wage growth, retaining zero wages next year.

$$g_{i,t+1} = \frac{2(y_{i,t+1} - y_{it})}{y_{i,t+1} + y_{it}}, \qquad y_{it} > 0, \quad y_{i,t+1} \geq 0.$$

y is annual nominal wages. When next-year wages are zero, $g = -2$.

2. Remove the age profile within the broader group G.

$$\epsilon_{i,t+1} = g_{i,t+1} - \widehat{m}_G(a_{it}), \qquad \widehat{m}_G(a) \text{ is a weighted cubic in age.}$$

The regression includes an intercept, so residuals average to zero within G.

3. Center within the finer age-band cell c.

$$\mu_c = \mathbb{E}_c[\epsilon], \qquad \text{deviation} = \epsilon - \mu_c.$$

A zero mean across all ages in G does not imply zero means in each age band. Semivariances use these within-cell deviations.

◀ Back ▶ Growth measure ▶ SIPP regression controls

Retirement contributions

Complementary evidence on participation in another saving vehicle. These contribution outcomes are not calibration targets.

Initial non-IRA stock ownership	Upside balance (pp)	SE
Nonholders	0.22	0.33
Holders	6.36***	2.19
Holder minus nonholder	6.14**	2.33

Outcome: any private IRA or annuity contribution, not the amount saved or the account's equity share.

Does the pattern extend to household purchases?

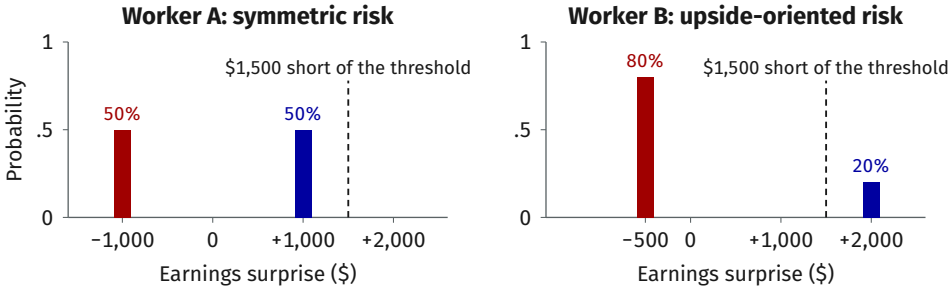
Complementary evidence on occasional purchases. Not used to fit the model.

Asinh dollar scale	Furnishings/ equipment $1,000	Selected nondurables/ services $10,000	Food $10,000
Upside balance, holding variance fixed			
Initial nonholders	0.007	0.001	0.002
	(0.013)	(0.008)	(0.005)
Initial holders	0.114***	0.012	0.009
	(0.041)	(0.020)	(0.017)
Holder minus nonholder	0.107**	0.011	0.006
	(0.046)	(0.024)	(0.019)
Baseline-weight-share average	0.027**	0.003	0.004
	(0.012)	(0.006)	(0.004)
Variance, holding upside balance fixed			
Initial holders	0.021***	0.003	0.001
	(0.006)	(0.004)	(0.003)
33,353 observations; 30 industry–education clusters			

Furnishings: positive among holders. Other baskets: imprecise.

▶ Why study consumption? ▶ Inference

Earnings gains near the threshold



Entry probability: **0%** for A **20%** for B

Earnings risk and stock market participation

Saving, entry opportunities and holding costs

The household must finance consumption, saving and any fee

M_t: current cash; c_t: consumption; A_t: saving carried forward.

$$c_t + A_t + D_t K_t = M_t, \qquad K_t = \kappa_R \xi_t.$$

D_t: stockholding indicator; K_t: fee offer; κ_R: mean fee; ξ_t: mean-one multiplier.

$$M_{t+1} = A_t\{R_{f,t+1} + \alpha_t(R^s_{t+1} - R_{f,t+1})\} + Y_{t+1}.$$

α_t: stock share; R_f, R^s: gross safe and stock returns; Y: disposable income.

- Positive consumption, no borrowing: $c_t > 0$, $A_t \geq 0$.
- Access $a_t \in \{0, 1\}$ is known before choice: $D_t \in \{0, 1\}$ and $D_t \leq a_t$.
- No short positions: $0 \leq \alpha_t \leq D_t$. Funded stocks: $\alpha_t A_t \geq \$1 \times D_t$.

◀ Back ▶ Access and fees ▶ Optimization

An investment opportunity permits entry; a fee makes it costly

d_t: prior stockholding; a_t: opportunity to invest now.

$$\Pr(a_t = 1 \mid d_t) = d_t + (1 - d_t)q.$$

- A previous nonholder receives a fresh opportunity with probability q.
- A previous holder can choose again. Exit returns the household to the nonholder opportunity process.
- An opportunity permits entry; it does not require participation.

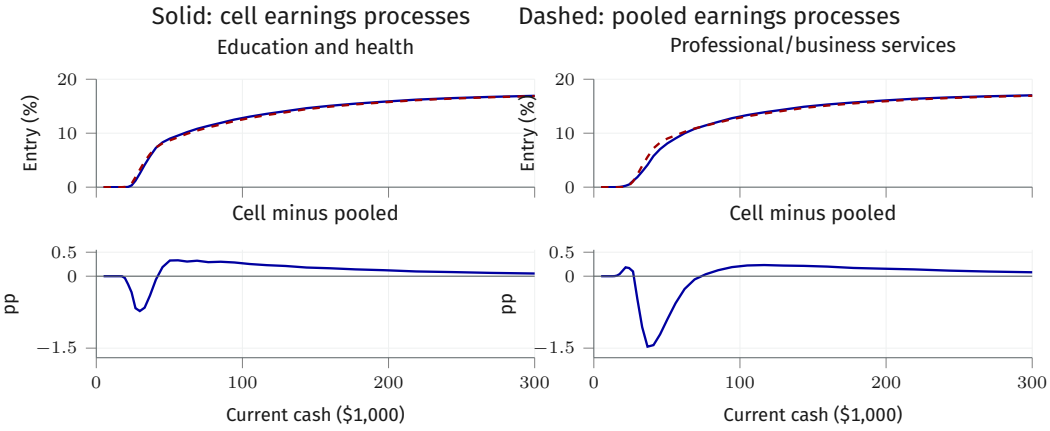
$$K_t = \kappa_R \xi_t, \qquad \mathbb{E}[\xi_t] = 1.$$

K_t: fee offer; κ_R: mean offer; ξ_t: independent lognormal multiplier.

The household pays the fee only if it chooses a funded stock position.

▶ Structural parameter estimates ▶ Offers versus payments ▶ Permanent access

Entry probabilities by current cash



Targeted financial moments

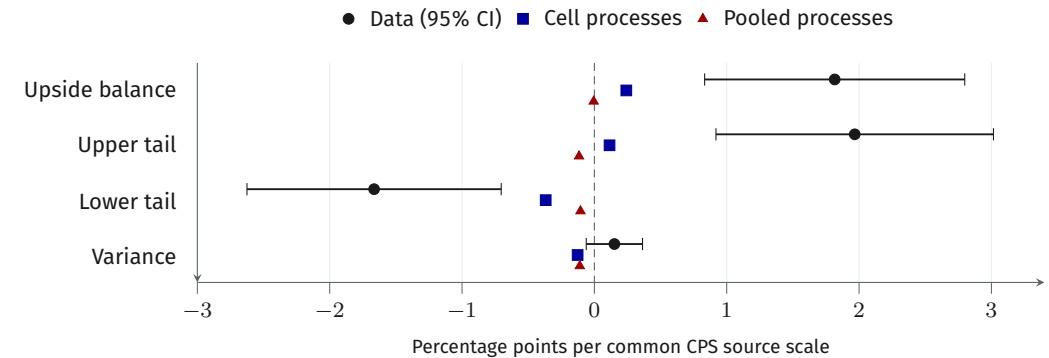
Moment	Unit	Data	SE	Model
Entry: low initial wealth	%	2.93	(0.26)	3.18
Entry: middle initial wealth	%	5.84	(0.36)	5.51
Entry: high initial wealth	%	10.36	(0.48)	10.44
Retention: low initial equity share	%	66.19	(1.43)	67.75
Retention: high initial equity share	%	76.31	(1.23)	74.34
Next equity share: all initial holders	%	33.38	(0.84)	34.40
Wealth change: initial nonholders	$	3,939	(778)	9,941
Wealth change: initial holders	$	1,946	(5,177)	511

Main discrepancy: excess nonholder wealth growth

◀ Back ▶ Additional diagnostics ▶ Alternative fit

Entry estimates from the model

17,606 matched initial nonholders, untargeted coefficients



Upper/lower rows are local coefficient transformations, not separate regressions.

◀ Back ▶ Permanent alternative

Earnings distributions in portfolio models

Study	Existing mechanism or comparison
Cocco, Gomes & Maenhout (2005)	Labor income, borrowing constraints and downside earnings risk
Catherine (2022)	Earnings downside when stock returns are poor
Shen (2024)	Skewed earnings risk, portfolio choice and participation costs
Gálvez & Paz-Pardo (2026)	Rich earnings dynamics and data/model risk-response regressions

This exercise: assess a specific prospective entry relationship after financial estimation.

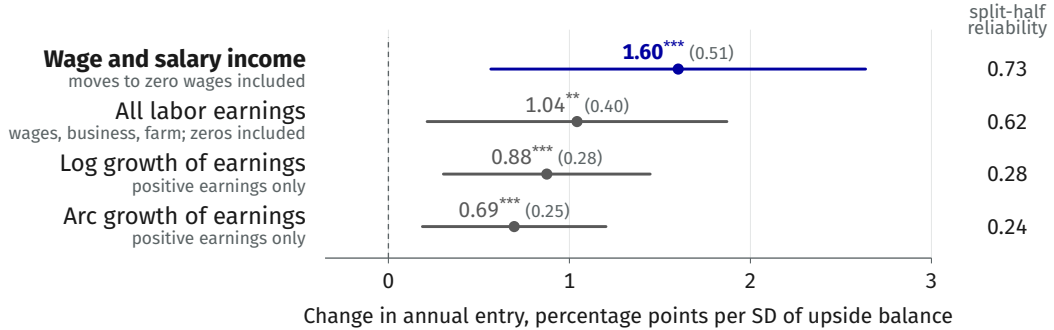
◀ Back ▶ Detailed comparison

Upside balance at fixed variance

	Symmetric risk	Upside-oriented risk
Lower outcome	$-1{,}000$ (50%)	-500 (80%)
Upper outcome	$+1{,}000$ (50%)	$+2{,}000$ (20%)
Lower partial variance	500,000	200,000
Upper partial variance	500,000	800,000
Total variance	1,000,000	1,000,000
Upside balance	**0**	**600,000**

Upside balance: upper minus lower partial variance.

Entry estimates across earnings measures



Positive entry estimates across all four measures. Zero-inclusive wages are the most reliable.

▶ Zeros and growth ▶ Skewness

Observed initial states

21,188 SIPP worker-year pairs, ages 25–63 at origin

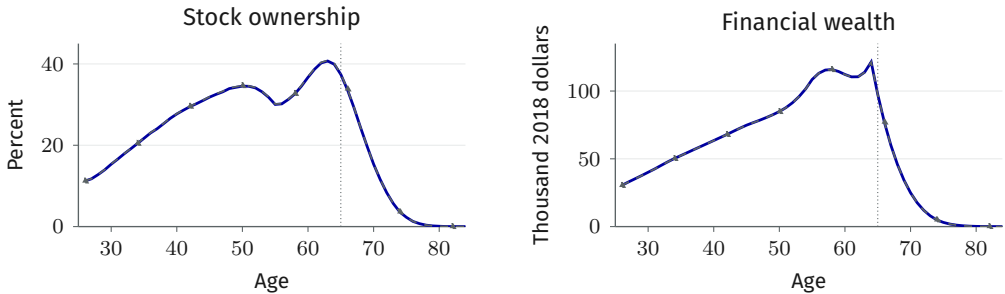
Input	Use in the model
Wages, wealth, ownership and shares	Latent earnings from wages; observed financial positions
Taxes and own pension contributions	Wages net of taxes and contributions
Own retirement income	PSID Social Security and pensions by education and age
Calendar returns	Returns on inherited portfolios in observed transitions

Future return expectations: stationary.

Conditional life-cycle profiles

Conditional projection from 452 observed age-25 origins in 29 groups

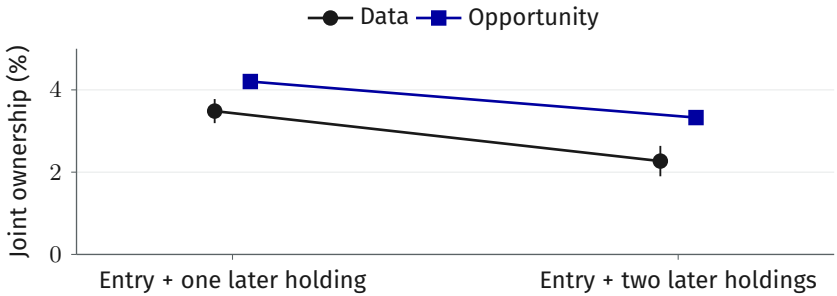
Solid: 141-node grid Dashed triangles: 201-node grid



Numerical accuracy: almost identical paths on finer cash and return grids.

Entry and subsequent ownership

Untargeted predictions on common data/model followup cohorts



Sustained ownership levels remain overpredicted.

▶ Common-cohort estimates

Samples

Sample	Observations
CPS linked earnings histories	88,862
CPS histories with zero wages the next year	5,490
SIPP full entry sample	27,419
SIPP valid-share sample at both dates	31,569
Model financial-fitting worker-year pairs	21,188
Matched model/data entry regression	17,606

The 31,569 allocation observations require positive broad and nonretirement financial assets and valid risky shares at both dates: 24,235 nonholders and 7,334 holders.

Worker transitions, household assets. Mixed-asset mutual funds included; retirement-account allocation is unobserved.

◀ Back ▶ Appendix index

Arc wage growth

$$g_{i,t+1} = \frac{2(y_{i,t+1} - y_{it})}{y_{i,t+1} + y_{it}}.$$

- Positive wages initially; zero wages the next year included: $g = -2$
- Remove a weighted cubic age profile within industry–education groups
- Then center within the finer industry–education–age-band cells
- Zero wages: nonemployment or a different income source, not necessarily dismissal

◀ Back ▶ Appendix index

Relation to skewness

	Upside balance C	Conventional skewness
Construction	Signed squared deviations	Cubed deviations, divided by SD^3
Interpretation	Upper minus lower contributions to variance	Standardized third moment
Extreme observations	Squared weighting	Cubed weighting

Both describe asymmetry. They are closely related in these data.

Why this measure? Additive variance decomposition; higher measured reliability (0.73 vs. 0.39).

These results do not show that upside balance predicts entry better than skewness.

◀ Back ▶ Saved skewness comparisons

Upside balance and skewness: the same controls

Zero-inclusive wage measure	Upside balance	Skewness
Each measure separately	1.601^{***}	0.912^{**}
	(0.506)	(0.414)
Both measures together	2.585	-0.726
	(1.670)	(1.195)
Split-half reliability	0.73	0.39

Same 27,419 observations; all headline controls, including V and $\sqrt{V}$.

Partial correlation given $V, \sqrt{V}$: 0.91. Neither joint coefficient is precisely separated.

Each measure uses its own source SD. CR1/t_{29} standard errors.

◀ Back ▶ Auxiliary total-earnings comparison

Alternative earnings measures

Construction	Upside balance (pp)	SE
Wages, zeros included	1.601***	0.506
Total labor earnings, zeros included	1.042**	0.405
Real log growth, positive support	0.875***	0.279
Nominal arc growth, positive support	0.695***	0.248

Controls: $V, \sqrt{V}, \mu, \mu^2$, household characteristics
Industry, education, age-band and year fixed effects

◀ Back ▶ Appendix index

Statistical inference

- **Coefficient tests:** 30 industry–education clusters, CR1/t_{29}
- **Wild-cluster bootstrap:** $p = 0.039$
- **Source uncertainty:** re-estimated CPS moments in separate checks
- Restricted samples: actual cluster count. Unadjusted coefficient stars.

Significant variance terms can give a small local slope

Holding upside balance and controls fixed, the raw-unit variance terms are

$$f(V) = b_V V + b_S \sqrt{V}, \qquad b_V < 0, \quad b_S > 0.$$

$$f'(V) = b_V + \frac{b_S}{2\sqrt{V}}.$$

A concave relationship: the fitted entry association first rises with variance, flattens, then falls. The two coefficients describe one curve.

Evaluation volatility	Lower quartile	Mean	Upper quartile
Local response (pp)	$+0.438^{***}$	$+0.104$	-0.212
	(0.135)	(0.116)	(0.136)

Full regression including C. Saved local responses per common source SD of C; full-covariance SEs. Raw coefficients are standardized coefficients divided by their respective source SDs. A small slope at the mean does not imply insignificant variance terms.

◀ Back ▶ Appendix index

Risk preferences and sorting

- Risk preferences affect participation thresholds and earnings responses
- Upside balance slopes may differ across preference groups
- Reported risk appetite: noisy sorting diagnostic

Sorting remains a possible source of the association.

Entry and subsequent ownership

Upside balance estimate (pp)	One followup	Two followups
Any entry	2.09***	3.31**
Entry and later holdings	0.98*	1.89**
Entry with an interruption	1.11**	1.42
Initial nonholders	9,236	3,254

Joint outcomes partition entry within each cohort.
Imprecise estimates of retention among entrants. Followup selection and weaker three-date reporting screen.

◀ Back ▶ Appendix index

Period differences and age groups

Young-worker SIPP entry gap (pp)	Estimate
Original period gap	2.885
After symmetric overlap weighting	2.623
Change in gap	-0.262
SE of change	0.294
p-value of change	0.381

PSID: all ages, 1999–2017 baselines: -0.18 (SE 0.39).
Ages 45–65, 2009–2017 baselines: 0.44 (SE 1.30). All-age evidence is inconclusive.

Exit among initial holders

A separate risk set: people who already own stocks.

Next-year exit among initial holders	Upside balance estimate
Exit coefficient	$+3.72$ pp
Clustered standard error	1.71 pp
Wild-cluster bootstrap p-value	0.12

The estimate points toward more exit, but inference is sensitive.

More marginal investors entering and later exiting is one possible explanation. This regression does not test that channel.

▶ Holder outcomes ▶ Selection and interpretation

Initial-holder outcomes

Initial-holder outcome	Upside balance (pp)	SE
Next-period ownership / retention	-3.72^{**}	1.71
Next risky share, controlling initial share	-0.51	0.94

- Retention wild-cluster bootstrap $p = 0.12$.
- All initial holders, including subsequent exits
- Initial-holder retention differs from survival among new entrants

◀ Back ▶ Appendix index

Why participation and average spending can move differently

Purpose of the expenditure evidence: compare the portfolio pattern with occasional purchases and regular spending. It is neither an entry robustness test nor a calibration target.

A simple benchmark: two dates, log utility, cash m, entry fee κ, saving and participation $d \in \{0, 1\}$.

$$c^*(m, d) = \frac{m - \kappa d}{1 + \beta}, \qquad \Delta\mathbb{E}[c] = -\frac{\kappa}{1 + \beta}\Delta \Pr(d = 1)$$

at fixed mean resources. More households can enter with only a small change in average consumption.

Empirical interpretation. Furnishings are occasional purchases; their positive primary estimate among holders does not establish a difference from other baskets. Dollar and original-construction estimates are imprecise.

The life-cycle model has one consumption good. It does not estimate separate durable-purchase decisions or total saving from IRA participation.

◀ Back ▶ Retirement contributions ▶ Index

The participation threshold follows from optimal saving

m: cash; c: consumption; $d \in \{0, 1\}$: entry; κ: fixed fee; β: discount factor.

With log utility, let $b > 0$ be the optimized gain in expected log portfolio return from stock access.

$$V_d(m) = \max_{0<c<m-\kappa d} \{\log c + \beta[\log(m - \kappa d - c) + bd]\}.$$

$$c_d(m) = \frac{m - \kappa d}{1 + \beta}, \qquad V_1 - V_0 = (1 + \beta) \log\left(1 - \frac{\kappa}{m}\right) + \beta b.$$

$$d^*(m) = 1 \iff m \geq m^* = \frac{\kappa}{1 - \exp[-\beta b/(1 + \beta)]}.$$

The fee absorbs a smaller share of cash when resources are higher.

This illustration has no terminal labor income. The life-cycle model adds future earnings uncertainty and repeated choices.

◀ Back ▶ Stationary selection formulas

A separate fixed-wealth illustration: who enters and stays?

Hold wealth w fixed. Draw earnings Y independently each date from distribution F.
κ: participation fee; h: safe terminal income; $b > 0$: optimized log-return gain.

$$w^* = \frac{\kappa}{1 - e^{-b}} - h, \qquad \pi(w, F) = \Pr(Y \geq w^* - w).$$

w^* is the required resource threshold; π is the participation probability at a given w, F.

$$\text{Entry} = \frac{\mathbb{E}[\pi(1-\pi)]}{\mathbb{E}[1-\pi]}, \qquad \text{Retention} = \frac{\mathbb{E}[\pi^2]}{\mathbb{E}[\pi]}.$$

$$\text{Entry + } k \text{ later holdings} = \frac{\mathbb{E}[(1-\pi)\pi^{k+1}]}{\mathbb{E}[1-\pi]}.$$

Expectations average across households with different w, F; k counts later holding dates.

These formulas isolate selection when wealth does not evolve.

The previous two-date model optimizes saving. This static illustration does not.

◀ Back ▶ Two-date mechanism ▶ Index

Why 58 cells? Why not 580?

Cells describe earnings environments; more cells require more information.

The 30 retained industry–education groups have two age bands: 60 possible group–age combinations. 58 have empirical support; two use the other age band in the model.

Support-only age-splitting trial	Current 58	Candidate 580
Median observations per cell	1,284	111
Median effective sample size	787	72
Cells passing existing support rule	58	50

Split each retained cell into ten narrower age groups. About 91% fail the existing support rule.

Finer cells need more data or estimates that share information across groups.

This checks data support only. No new earnings processes or portfolio estimates were fitted.

◀ Back ▶ Model mapping

Full entry projections: upside balance and local tail contrasts

Upper/lower rows are local algebraic contrasts, not separate regressions.

Entry estimate (pp)	Data	SE	Cell	Pooled	Net
Upside balance	1.816^{***}	(0.480)	0.241	-0.005	0.247
Upper tail	1.967^{***}	(0.513)	0.115	-0.116	0.230
Lower tail	-1.665^{***}	(0.470)	-0.368	-0.105	-0.263
Variance	0.151	(0.104)	-0.127	-0.110	-0.016

- Pooled: both tail estimates negative. Cell: upper positive, lower negative.
- Neither relationship is targeted. Both cell estimates remain much smaller than in the data.

▶ Process and policy contributions ▶ Coefficient figure ▶ Why the test is informative

Selected references

- Catherine (2022). Countercyclical Labor Income Risk and Portfolio Choices over the Life Cycle. *Review of Financial Studies*.
- Catherine, Sodini & Zhang (2024). Countercyclical Income Risk and Portfolio Choices: Evidence from Sweden. *Journal of Finance*.
- Shen (2024). Countercyclical Risks, Consumption, and Portfolio Choice: Theory and Evidence. *Management Science*.
- Bonaparte, Korniotis, Kumar, Michaelides & Zhang (2025). Stock Market Ownership Transitions. *Management Science*.
- Gálvez & Paz-Pardo (2026). Richer earnings dynamics, consumption and portfolio choice over the life cycle. *Journal of Financial Economics*.
- Briggs, Cesarini, Lindqvist & Östling (2021). Windfall gains and stock market participation. *Journal of Financial Economics*.

Closest studies and this paper's contribution

Study	Empirical object
Catherine, Sodini & Zhang (2024)	Current risky shares including zeros, with participation decomposition. Unconditional and cyclical earnings skewness.
Shen (2024)	Changes in risky shares in the stockholder sample, related jointly to skewness and variance.
Bonaparte et al. (2025)	Actual ownership transitions related to income and wealth changes.
Gálvez (2023, working paper)	Participation responses following persistent earnings shocks.

Here: earlier CPS earnings upside balance and later SIPP entry among initial nonholders, at similar variance and resources.

Entry can include former owners returning to the market.

◀ Back ▶ Risk measures ▶ Index

Permanent participation eligibility

$$T_i \in \{0, 1\}, \qquad T_i \text{ remains fixed throughout the path.}$$

- Current holders and reported former holders: $T_i = 1$.
- Among other initial nonholders, $\Pr(T_i = 1) = 0.119$.
- Eligible: annual choice to pay and hold equity. Ineligible: safe asset only.
- Five parameters fitted separately to the same eight financial targets.

Permanent exclusion applies to a fixed share of initially unknown nonholders.

Financial fit under both access models

Moment	Data	Opportunity	Permanent
Entry: low initial wealth	2.93	3.18	3.21
Entry: middle initial wealth	5.84	5.51	5.49
Entry: high initial wealth	10.36	10.44	10.39
Retention: low initial equity share	66.19	67.75	68.01
Retention: high initial equity share	76.31	74.34	73.83
Next equity share: all initial holders	33.38	34.40	33.81
Wealth change: initial nonholders	3,939	9,941	10,005
Wealth change: initial holders	1,946	511	501

Both models: positive wealth growth, excessive nonholder accumulation.

Entry estimates under permanent eligibility

Entry estimate (pp)	Data	Opportunity net	Permanent net
Upside balance	1.816	0.247	0.171
Upper tail	1.967	0.230	0.155
Lower tail	-1.665	-0.263	-0.188
Variance	0.151	-0.016	-0.016

- Upside balance: pooled 0.211 pp, cell 0.382 pp. A positive association can arise with pooled processes.
- Cells strengthen the association here too. Magnitudes remain well below the data.

Upper/lower rows are local coefficient transformations, not separate regressions.

Participation dynamics under both access models

Untargeted outcome	Data	Opportunity	Permanent
Re-entry: reported former holders (%)	27.16	9.83	57.68
Entry + one later holding (%)	3.48	4.20	3.82
Entry + two later holdings (%)	2.27	3.33	2.24
Projected age-45 ownership (%)	—	31.83	16.31
Windfall response after two years (pp)	—	15.25	3.44

Permanent eligibility: closer fit to longer sustained ownership; excessive re-entry among reported former holders.

Parameter estimates by access model

	Opportunity	Permanent
Risk aversion γ	8.801	8.917
Discount factor β	0.811	0.810
Recurring mean κ_R	\$524	\$413
Log-cost standard deviation σ_κ	0.944	1.013
Opportunity q / active share p	0.176	0.119
EIS ψ, fixed	0.5	0.5
Additional entry fee	0	0

Parameters fixed before earnings-risk and dynamic comparisons. No binding bounds.

Estimation criterion and grid refinement

$$Q = Q_E + 0.1Q_H + Q_W = 9.77.$$

- Marginal covariance blocks: entry, holder outcomes and wealth changes.
- Wealth covariance: \$2,000 regularization. Primary-grid growth buffer: \$500 for both groups.
- Native convergence. At 201/7: wealth growth of \$9,937 for nonholders and \$498 for holders.
- Excluded targets: earnings risk, duration, windfalls and life-cycle profiles.

◀ Back ▶ Other numerical diagnostics ▶ Index

Opportunity friction and recurring payments

- Opportunity: a nonholder can reconsider participation.
- Advice, attention and search are possible interpretations, not separately identified.
- Recurring cost paid only when holding equity.
- Offered cost: mean $524, median $335. Participants select lower draws.

Selected recurring payment	Real 2018 dollars
Entrants, supported origins	235
Retained holders	359

Wealth growth and saving net of returns

$$\Delta A_t = \text{gains on inherited assets} + Y_t - c_t - D_t \kappa_R \xi_t.$$

Nonholder wage tercile	Data growth ($)	Model growth ($)
Low	2,388	3,588
Middle	236	8,705
High	9,195	17,535

Holders have positive wealth growth but negative aggregate saving net of returns.

Pooled processes and initial-state sorting

- Fixed: origins, group wage levels, earnings posteriors and calendar returns.
- Remove cell adjustments; reoptimize at common financial parameters.
- Initial resources may remain correlated with earnings cells.
- Pooled holder wealth growth: about $483 below zero. Positive growth is imposed in the fitted cell model.

Four-corner accounting by risk measure

c: cell process; p: pooled process; B: entry-regression coefficient.
$B(\text{process}, \text{rule})$: next-year outcome process crossed with a previously solved rule.

$$\Delta_L = \tfrac{1}{2}\{B(c,c) - B(p,c) + B(c,p) - B(p,p)\},$$
$$\Delta_P = \tfrac{1}{2}\{B(c,c) - B(c,p) + B(p,c) - B(p,p)\}.$$

Entry estimate (pp)	Processes	Decisions	Total
Upside balance	+0.274	-0.027	+0.247
Upper tail	+0.302	-0.072	+0.230
Lower tail	-0.245	-0.018	-0.263
Variance	+0.029	-0.045	-0.016

Decision rules dampen the upper-tail contribution and reinforce the negative lower-tail contribution.

One-year accounting from common observed origins. The crossed rule retains its own later expectations. Outcomes include cash, work, persistent states and sample selection.

Entry and later ownership: estimates

	One later holding	Two later holdings
Origins	5,459	1,801
Data upside balance (pp)	0.11	0.77
Data SE	(0.66)	(0.77)
Opportunity net, two seeds	0.19–0.20	0.18–0.22
Permanent net, two seeds	0.14–0.19	0.08–0.13

Smaller common cohorts: imprecise empirical upside balance estimates. Simulation-seed ranges are not confidence intervals.

Model fit and remaining discrepancies

Comparison	Model result	Discrepancy
Entry by wealth	Increasing across wealth terciles	Imperfect financial fit
Entry and earnings risk	Opposing net tail signs	Magnitudes well below data
Participation dynamics	Entry rises after gains and windfalls	Sustained levels too high; re-entry too low
Saving and positions	Positive wealth changes for both groups	Excess nonholder accumulation; too few small accounts

Permanent alternative: excessive re-entry; closer sustained-ownership fit.

Earnings asymmetry and joint downturn risk

Study	Relevant finding or comparison
Catherine (2022)	Earnings skewness varies with market conditions. Removing that cyclicality preserves earnings variance, persistence and ordinary earnings–return covariance.
Gálvez & Paz-Pardo (2026)	Compares an untargeted income-risk/risky-share regression in data and financially estimated models.
This paper	Earnings upside balance predicts subsequent entry. The model's resources mechanism operates with independent future stock returns.

Upside balance measures upside-minus-downside partial variance, rather than conventional third-moment skewness or dependence on market returns.

Cell/pooled model predictions change the full earnings process, including mean growth. They do not isolate skewness alone.

◀ Back ▶ Comparison scope

Household optimization: states, choices and constraints

Current state	t: age; z: persistent earnings and wage-work states; d: prior stockholding
Resources and fee	m: cash before the fee; $K = \kappa_R \xi$: fee offer; $x = m - jK$: cash after payment
Current choices	$j \in \{0, 1\}$: stockholding; A: saving; α: stock share; consumption $c = x - A$
Next resources	$R(\alpha)$: gross portfolio return; Y': disposable income; primes denote next year

Choose consumption, saving and stocks to maximize lifetime Epstein–Zin utility, subject to

$$c + A + jK = m, \quad c > 0, \quad A \geq 0, \quad 0 \leq \alpha \leq j, \quad \alpha A \geq \$1\, j.$$

β: discount factor; γ: risk aversion; ψ: elasticity of intertemporal substitution.

s_t: survival; q: nonholder access probability; κ_R: mean fee offer; ξ: mean-one multiplier.

▶ Value functions ◀ Back ▶ Access and fees

Where future earnings enter optimal decisions

Let $h = 1 - 1/\psi$ and $\omega = 1 - \gamma$. Process L gives future earnings probabilities.
V^L_{jt}: value conditional on today's stockholding choice j and cash after fees x.

$$V^L_{jt}(z,x) = \max_{(A,\alpha)\in\mathcal{A}_j(x)} \left[(x-A)^h + \beta s_t \left\{\mathbb{E}_L\left[\bar{V}^L_{t+1}(z', j, AR(\alpha)+Y')^\omega \mid z\right]\right\}^{h/\omega}\right]^{1/h}.$$

With access and fee offer $\kappa_R\xi$, choose the better stockholding option:

$$W^L_t(z,m,\xi) = \max\{V^L_{0t}(z,m), V^L_{1t}(z, m-\kappa_R\xi)\}.$$

$\bar{V}^L$: value before access and fee draws; $q_d = d + (1-d)q$.

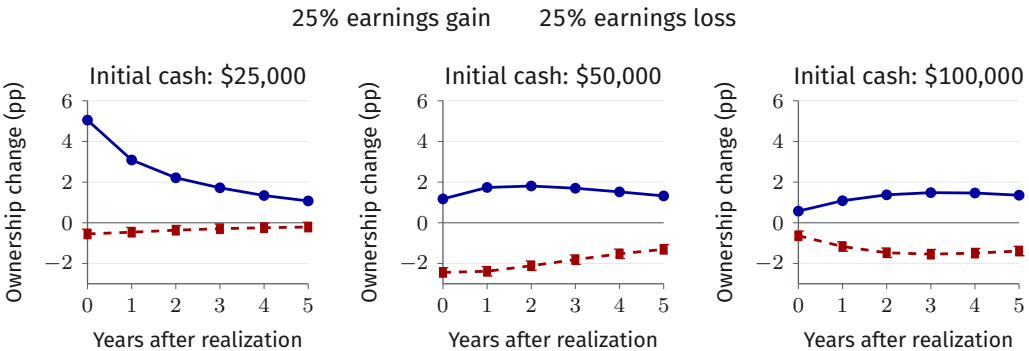
$$\bar{V}^L_t(z,d,m)^\omega = (1-q_d)V^L_{0t}(z,m)^\omega + q_d\mathbb{E}_\xi[W^L_t(z,m,\xi)^\omega].$$

$\mathcal{A}_j(x)$ contains the constraints on the previous slide. At age 84, $\bar{V}^L_{84} = m$.
The earnings process changes expected future utility, and hence today's saving and entry.

◀ Back ▶ Economic interpretation ▶ Index

Response to realized earnings shocks

Age 45: a one-year shock equal to 25% of a fixed reference gross wage



► Earnings news at fixed current resources ◄ Back

Realized earnings and earnings news

Experiment	What changes	Interpretation
Realized transitory shock	Current cash; fixed future processes	Gains can finance entry
Upper-tail news	Future distribution; fixed current resources	Participation response can be small
Upside balance news	Future distribution; fixed current resources	Response varies with cash and horizon

Upside balance-news responses can reverse sign at low cash. More upside need not raise entry.

Numerical precision

- Financial fit, matched entry projections and four-corner accounting: declared finer-grid checks passed.
- Life-cycle grids: 141/5 and 201/7. Some local policies, cash responses and consumption remain grid-sensitive.
- Realized-earnings events: separate 301-node refinement.
- Simulation variance and grid differences measure numerical, not sampling, uncertainty.

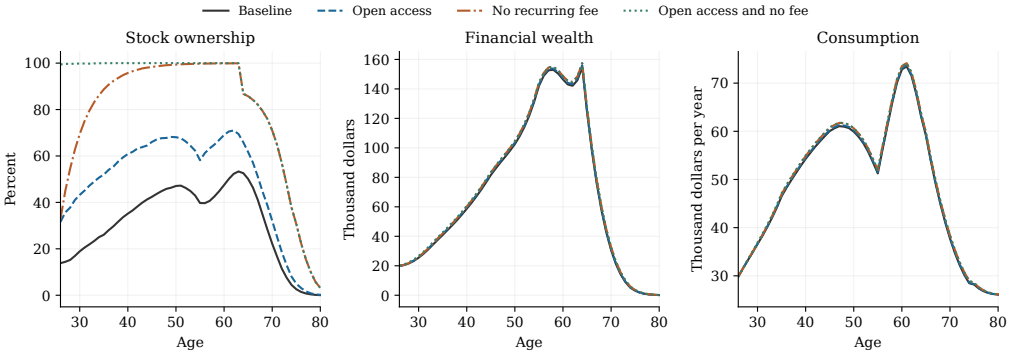
Additional model experiments

Family	Exercises in both specifications
Saving and portfolios	Age profiles, consumption, funded shares, entry and exit
Access frictions	Open access, zero and altered recurring costs
Earnings and retirement	Transitory and persistent risk, income insurance, pensions
Preferences and returns	Risk aversion, patience, EIS, expected returns and volatility
Household responses	Realized earnings, windfalls, fixed-resource news, value of future income

39 scenarios per model, plus conditional policy and valuation diagnostics. Alternative and sensitivity results in the online appendix.

Participation-friction counterfactuals

Remove each friction; reoptimize household choices



SMM targets and weighting

$$\theta = (\gamma, \beta, \kappa_R, \sigma_\kappa, q), \qquad \widehat{\theta} = \arg\min_{\theta} \{Q_E(\theta) + 0.1 Q_H(\theta) + Q_W(\theta)\}.$$

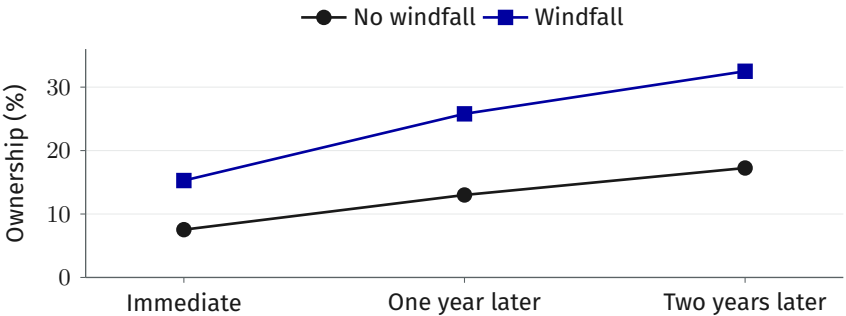
Block	Targets	Economic discipline
Entry (3), Q_E	Three rates by initial wealth	Opportunities and affordability
Holders (3), Q_H	Two retention rates and one next equity share	Continued participation and allocation
Wealth (2), Q_W	Two changes by initial ownership	Saving and resource use

Weight 0.1: entry and wealth growth receive priority over retention and share fit.

◀ Back ▶ Financial fit ▶ All eight targets ▶ Exact criterion

Windfall response

Untargeted experiment: $150,000 to 423 age-44 initial nonholders



Model: $+7.7$ pp immediately; $+15.2$ pp after two years.
Swedish lottery evidence: about $+12$ pp in the winning year (Briggs et al., 2021).

◀ Back ▶ Realized earnings gains ▶ Index

Full entry specification: initial nonholders only

Estimate only among people with no stock or mutual-fund holdings at t.

$$
\begin{aligned}
\mathbf{1}\{\text{Participates}_{i,t+1}\} = \alpha &+ \beta_E \widetilde{C}_{c(i,t)} + \gamma_1 \widetilde{V}_{c(i,t)} + \gamma_2 \widetilde{\sqrt{V}}_{c(i,t)} \\
&+ \gamma_3 \widetilde{\mu}_{c(i,t)} + \gamma_4 \left(\widetilde{\mu}_{c(i,t)}\right)^2 \\
&+ \delta_1 \log\Big(\max\{A_{it}, 1\}\Big) + \delta_2 \log\Big(\max\{\bar{y}^{(1)}_{it} + 1, 1\}\Big) \\
&+ \delta_3 \mathbf{1}\Big\{\min_{m \in \mathcal{M}_{it}} y^{(1)}_{itm} \le 0\Big\} + \delta_4 a_{it} + \delta_5 a_{it}^2 + \delta_6 \text{Male}_{it} \\
&+ \eta_{j(i,t)} + \kappa_{e(i,t)} + \lambda_{b(i,t)} + \tau_{t+1} + u_{i,t+1}.
\end{aligned}
$$

Earnings measures: C upside balance; V variance; μ mean adjusted growth. Tildes use fixed CPS source scales; c is the baseline earnings cell.

Resources: A gross assets; $y^{(1)}$ first-listed-job earnings; $\bar{y}^{(1)}$ its mean over eligible months $\mathcal{M}$. The indicator records any month with nonpositive earnings. a is age.

Fixed effects: j industry, e education, b age band, plus outcome year. Participation is positive ownership in any eligible month of the year.

◀ Back ▶ Measurement versus controls ▶ Inference

Does the entry association depend on the earnings measure?

Growth measure	(1) Wages zeros included Arc	(2) Total earnings zeros included Arc	(3) Total earnings positive only Real log	(4) Total earnings positive only Arc
Upside balance C	1.601^{***}	1.042^{**}	0.875^{***}	0.695^{***}
	(0.506)	(0.405)	(0.279)	(0.248)
Split-half reliability of C	0.73	0.62	0.28	0.24
Age profile removed in CPS	Yes	Yes	Yes	Yes
V, $\sqrt{V}$, mean and mean^2	Yes	Yes	Yes	Yes
Resources and demographics	Yes	Yes	Yes	Yes
Industry, education, age band, year effects	Yes	Yes	Yes	Yes

Every column: 27,419 nonholder transitions; mean entry 8.82%; 58 assignment cells and 30 inference groups.

Reliability asks whether two random halves of CPS workers give similar cell measures; values use the Spearman–Brown adjustment. Removing the age profile constructs the CPS measure; it is not a second cubic control in SIPP.

◀ Back ▶ Variance coefficients ▶ Controls ▶ Inference

Additional specification results

Specification	Upside balance (pp)	SE
Baseline: valid shares at both dates	1.65**	0.60
Industry × year; education × year	1.66**	0.62
Industry × education effects	1.86**	0.88
Reported, self-reported, stable households	2.56***	0.56
Controls for realized earnings	1.57**	0.59

The first three rows use 24,235 initial nonholders with positive broad and nonretirement financial assets and valid shares at both dates. Reporting screens and realized-earnings controls use their own samples.

▶ Risk preferences and sorting ▶ Exact samples

How the literature uses earnings variance

- Expected-utility models generally depend on the full distribution.
- Empirical studies use variance measures of subjective uncertainty, persistent risk or transitory risk.
- Some calibrated models impose Gaussian earnings innovations; their parameters summarize a restricted distribution.
- Recent portfolio models explicitly allow skewness and nonlinear earnings dynamics.

This paper asks an empirical question: what does the balance of favorable and adverse variation predict about entry, given variance?

Which comparisons are performed?

Comparison	Scope
Upside balance and skewness	Closely related. Saved joint regressions do not establish incremental information beyond skewness.
Cell and pooled earnings	Same model and financial parameters; optimal rules recomputed. This is the main earnings-environment comparison.
Opportunity and permanent access	Separately fitted to the same financial targets; different dynamic misses
Published structural models	Mechanism comparison. No common-data re-estimation of entire published models.

◀ Back ▶ Skewness horse race ▶ Access models

Auxiliary total-earnings horse race

Saved zero-inclusive total-earnings comparison	Result
Upside balance coefficient (pp)	3.782
Two-survey standard error (pp)	2.261
Upside balance p-value	0.100
Two-test adjusted q-value	0.200
Joint test: both slopes zero	$p = 0.038$
Joint test: broader dependence sensitivity	$p = 0.145$

Total earnings, including zeros; adds zero-wage probability. Separate two-survey inference.

This is not the headline wage regression. The evidence does not establish an incremental association beyond skewness.

Measurement adjustment and regression controls

Step	Purpose
CPS earnings construction	Remove an industry–education-specific age profile from growth, then calculate within-cell distributional moments.
SIPP entry regression	Condition participation on individual resources, age and sex, plus industry, education, age-band and year fixed effects.

The CPS age cubic is a measurement step, not a repeated SIPP control. Individual age and group effects in SIPP adjust the participation comparison.

Sex is an individual participation control; the headline risk cells are not sex-specific.

Separate industry, education and age-band effects do not absorb every industry–education–age interaction.

◀ Back ▶ Construction ▶ Full regression

The coefficient's unit

$$\widetilde{C}_c = \frac{C_c - \bar{C}_{\mathrm{CPS}}}{s_{C,\mathrm{CPS}}}, \qquad s_{C,\mathrm{CPS}} \simeq 0.034.$$

A one-unit increase in $\widetilde{C}$ means a 0.034 increase in raw upside balance.

The fitted coefficient is 0.01601 in probability units, or 1.601 percentage points.

It uses the same variance and individual controls throughout.

This unit describes differences between earnings environments. It is not a 1% earnings rise or an individual windfall.

◀ Back ▶ Tail units ▶ Illustration

Why different questions use different samples

Question	Observations	Reason
Main SIPP entry	27,419	Initial nonholders with required annual outcomes and controls
Allocation checks	24,235	Initial nonholders with positive financial denominators and valid shares at both dates
Entry plus later holdings	9,236 / 3,254	One / two additional complete followups
Model financial fit	21,188	Matched usable model states, including initial holders
Model entry regression	17,606	Initial nonholders within the model sample
SCE reallocation / share	2,136 / 3,332	Different outcome availability and eligibility

Model and data coefficients are compared on the same 17,606 origins. The 27,419-observation headline uses the broader entry sample.

◀ Back ▶ Additional samples

Which ownership sequences enter the sample?

Initial nonholders with observed ownership in every required follow-up year.

Four-year sequence	Entry	Entry + two later holdings
$0, 1, 1, 1$	1	1
$0, 1, 0, 1$	1	0
$0, 0, 1, 1$	0	0
$1, 0, 1, 1$	Outside the initial-nonholder sample	

Each year defines a new baseline. A later $0, 1, 1$ window can enter the shorter-horizon cohort.

Missing required follow-up excludes the observation. Ownership means any stockholding during that year.

Retention among entrants is a separate outcome; its estimates are imprecise.

Why probabilities do not mechanically shrink slopes

Y: observed entry indicators; X: regressors; W: survey-weight matrix.

$\widehat{\beta}$: weighted regression coefficients; p: predicted entry probabilities.

$$\widehat{\beta}(Y) = (X'WX)^{-1}X'WY.$$

If $p = \mathbb{E}[Y \mid X]$, then

$$\mathbb{E}[\widehat{\beta}(Y) \mid X] = \widehat{\beta}(p).$$

Averaging binary realizations into probabilities removes simulation noise; it does not mechanically reduce the expected regression coefficient.

Current matched mean: model 6.375%, data 6.375%. No exact-zero model probabilities among 17,606 origins.

The shortfall concerns how probabilities vary with earnings upside balance. Model probabilities need not equal the true conditional expectation.

◀ Back ▶ Coefficient comparison

Completed alternatives do not close the magnitude gap

Specification	Upside balance (pp)	Nonholder wealth error	Holder wealth error
Current cell model	0.241	Reference	Reference
Cash-dependent opportunities	0.337	+$157	+$11
$50 recurring cost + setup	0.267	+$302	+$10
$100 recurring cost + setup	0.253	+$362	+$11
Matched empirical coefficient	1.816		

Wealth columns: change in each group's absolute wealth-growth error; positive is worse. Same 141-node resolution. Both groups also worsen on the finer grid.

Low-fee versions predict retention above 96%. No tested specification resolves magnitude while preserving the requested wealth fit.

◀ Back ▶ Permanent alternative

The earnings process: states and wage levels

$$F^{L}_{g,t}(\eta_{t+1}, n_{t+1}, W_{t+1} \mid \eta_t, n_t), \qquad L \in \{\text{cell}, \text{pooled}\}.$$
$$W_{gt} = \mathbf{1}\{n_t = 0\} \exp\{f_h(t) + \ell_g + B_g(t) + \eta_t + \varepsilon_t\}.$$

State / component	Meaning
n_t	0 with wage work, 1 without wage work
$h,\ g$	Education type, industry–education group
$f_h(t),\ \ell_g$	Education age profile and group wage level
$B_g(t),\ \eta_t,\ \varepsilon_t$	Cell growth profile, persistent state, transitory shock

Gross wages W_{gt} enter disposable resources Y_t after taxes and employee contributions. Other resources can remain when wages are zero.

◀ Back ▶ Cell coefficients ▶ Career transitions ▶ Cell and pooled

The cell adjustment vector and its joint targets

Coefficient	What changes
a_c^{exit}	Wage-work departure hazard
b_c	Deterministic age-growth profile
s_c	Transitory and reporting-error scale
a_c	Transitory probabilities and persistent jump hazards

$$B_g(t) = \begin{cases} b_{g0}(t-35), & t < 45, \\ 10b_{g0} + b_{g1}(t-45), & t \geq 45. \end{cases}$$

Fit the four coefficients jointly to the zero-wage share, mean arc growth, and upper/lower partial variances.

The persistent versus transitory allocation of asymmetry is weakly identified. Pooling sets all four adjustments to zero and reoptimizes choices.

◀ Back ▶ Persistent transitions ▶ Temporary shocks

Persistent earnings and wage-work transitions

$$\eta_{t+1} = \rho_h \eta_t + \zeta_{t+1} + J_h^+ I_{t+1}^+ - J_{h,b}^- I_{t+1}^- - m_{g,b}, \quad \zeta_{t+1} \sim N(0, s_{0h}^2).$$

ρ_h: persistence. $m_{g,b}$: centering drift. Promotion, loss and no jump are mutually exclusive outcomes.

$$\widetilde{p}_{c,b}^{\pm} = \min\{\widehat{p}_{c,b}^{\pm}, .45\}, \qquad p_{c,b}^{\pm}(\eta) = \min\{\widetilde{p}_{c,b}^{\pm} e^{\pm k_h^{\pm} \eta/\sigma_{\eta,h}}, .45\}.$$

$\widehat{p}^{\pm}$ are the tilted probabilities before caps, defined on the linked normalization slide.

$$\Pr(n_{t+1} = 1 \mid n_t = 0, \eta_t) = \min\{\delta_{h,b} e^{a_c^{\text{exit}}} e^{-\phi_h \eta_t/\bar{\eta}_h}, .6\},$$

$$\Pr(n_{t+1} = 0 \mid n_t = 1) = \min\{r_{h,b}^0 e^{r_h}, .95\}.$$

$n = 0$: wage work. $n = 1$: no wage work. Earnings states evolve in both states. Wage-work return r_h differs from investment opportunity q.

◀ Back ▶ Tilt and normalization ▶ Parameters

What does the persistent-shock normalization hold fixed?

Before caps and state dependence, cell c tilts promotion/loss probabilities:

$$\widehat{p}^{+}_{c,b} = \chi_{c,b} p^{+}_{h,b} e^{a_c}, \qquad \widehat{p}^{-}_{c,b} = \chi_{c,b} p^{-}_{h,b} e^{-a_c},$$

$$\chi_{c,b} = \frac{p^{+}_{h,b}(J^{+}_{h})^2 + p^{-}_{h,b}(J^{-}_{h,b})^2}{p^{+}_{h,b} e^{a_c}(J^{+}_{h})^2 + p^{-}_{h,b} e^{-a_c}(J^{-}_{h,b})^2}.$$

$$\widehat{p}^{+}_{c,b}(J^{+}_{h})^2 + \widehat{p}^{-}_{c,b}(J^{-}_{h,b})^2 = p^{+}_{h,b}(J^{+}_{h})^2 + p^{-}_{h,b}(J^{-}_{h,b})^2.$$

This fixes a raw jump second moment before later adjustments.

Probability caps and state dependence follow. Centering, employment transitions and temporary shocks also matter for total earnings variance and growth.

◀ Back ▶ Complete transition ▶ Temporary component

Transitory earnings and reporting error

Transitory earnings

$$\varepsilon^0 \sim (1-p_2)N(0,s_1^2) + p_2 N(m_2, s_2^2).$$

Ten raw nodes e_j with probabilities p_{0j}. Cell tilt and scale give

$$p_{cj} \propto p_{0j} \exp\{a_c \operatorname{sgn}(x_j) x_j^2\}, \quad x_j = \operatorname{clip}\big((e_j - E_0 e)/sd_0(e), -4, 4\big),$$

$$\varepsilon_{cj} = e^{s_c} e_j - \log \sum_k p_{ck} e^{e^{s_c} e_k} + \log \sum_k p_{0k} e^{e_k}.$$

The location adjustment preserves $E[e^{\varepsilon}]$, the baseline wage multiplier. It does not fix variance.

Reporting error

$$\log W_t^{\text{obs}} = \log W_t + u_t \quad (W_t > 0), \qquad u_t^0 \sim (1-p_u)N(0,t_1^2) + p_u N(0,t_2^2).$$

Cell scale also multiplies reporting-error nodes by e^{s_c}. Reporting error changes measurement, not household resources.

◀ Back ▶ Shock parameters ▶ Cell coefficients

Earnings parameters: career dynamics

Parameter	Noncollege	College
Persistence ρ_h	0.946	0.944
Small innovation SD s_{0h}	0.024	0.037
Promotion size J_h^+	0.328	0.522
Loss size J_h^-, ages 25–54	0.742	0.419
Loss size J_h^-, ages 55–64	1.150	1.346
Promotion state sensitivity k_h^+	0.248	0.140
Loss state sensitivity k_h^-	0.148	0.931
Age-band order: 25–34, 35–44, 45–54, 55–64		
Base promotion probabilities $p_{h,b}^+$	.117, .039, .013, .046	.082, .019, .009, .021
Base loss probabilities $p_{h,b}^-$	.036, .032, .023, .035	.010, .029, .030, .043
$f_h(t) = a_{0h} + a_{1h}x + a_{2h}x^2 + a_{3h}x^3, \quad x = (t-45)/20$		
a_{0h}, a_{1h}	10.502, .202	11.153, .118
a_{2h}, a_{3h}	$-.239, -.004$	$-.361, .087$

Jump sizes in log points; annual probabilities before cell and state adjustments.

◀ Back ▶ Other parameters ▶ Main earnings slide

Earnings parameters: shocks and wage work

Parameter	Noncollege	College
Transitory large-component probability p_2	.044	.031
Transitory small-component SD s_1	.060	.101
Transitory large-component mean m_2	$-.339$	$-.349$
Transitory large-component SD s_2	.627	.620
Reporting large-component probability p_u	.230	.231
Reporting small-component SD t_1	.164	.182
Reporting large-component SD t_2	.913	.819
Departure state sensitivity ϕ_h	1.463	1.435
Wage-work return log shift r_h	1.926	1.996
Age-band order: 25–34, 35–44, 45–54, 55–64		
Base departure probabilities $\delta_{h,b}$	.074, .068, .057, .073	.039, .030, .041, .069
Return inputs $r^0_{h,b}$	.244, .170, .084, .047	.924, .292, .263, .094

Log-point shocks. Return inputs precede the fitted shift and probability cap.

◀ Back ▶ Career parameters ▶ Process definition ▶ Main

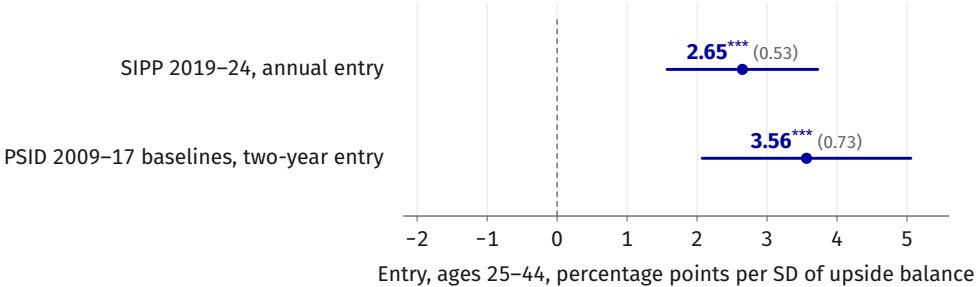
Why measure earnings risk earlier?

- CPS earnings changes are measured before the SIPP investment outcomes.
- A current shock cannot directly enter both the historical risk measure and the current portfolio report.
- Persistent group conditions, worker sorting and correlated shocks can still connect the two periods.

The timing reduces a source of contemporaneous correlation. It does not establish causal assignment of earnings risk.

◀ Back ▶ Group controls

Entry at ages 25–44



▶ Period gap and all-age results

Why earnings targets do not guarantee entry signs

1. Fit wages and wage-work transitions to earnings data.
2. Financial targets discipline saving, participation costs and access.
3. Neither step targets how entry varies with earnings upside balance.

The model must convert those earnings distributions into the observed entry response through resources and choices.

Matching the earnings moments alone does not guarantee the sign or size of that response.

◀ Back ▶ Fitting targets ▶ Mechanisms

How large is the entry difference?

Study and outcome	Risk comparison	Difference
This paper: next-year entry among nonholders	1 source SD in upside balance	$+1.60$ pp
Catherine, Sodini & Zhang (2024): current participation	2 SD in cyclical skewness	-2.3 to -8.5 pp

Here: 1.60 points relative to an 8.82% mean entry rate.

About 16 more entrants per 1,000; 18% of the mean, not 18 percentage points.

Different risk measures, populations and outcomes. The published range is across specifications, not a confidence interval.

Source: Catherine, Sodini & Zhang (2024), Table IV and p. 1774.

Related work in progress

van den Akker, de Bresser, Teppa & Trucchi (2026)

Subjective Expectations and Household Portfolio Choices

- Subjective expectations linked to administrative portfolios.
- Abstract: upward income risk predicts investment; entry patterns are concentrated among younger households.

Here: measured group earnings upside balance and entry, controlling for total variance and resources.

Related conference work, verified in the IAAE 2026 program. Full specifications and journal publication have not been verified.

Two-period illustration: the optimization problem

Choose consumption c_0, safe saving s, stock investment x and entry $d \in \{0, 1\}$:

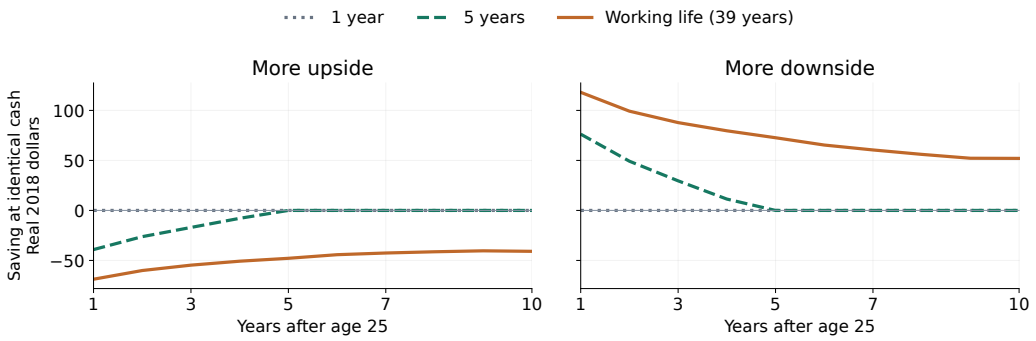
$$\max_{s,x,d} \log c_0 + \mathbb{E} \log c_1, \qquad c_0 = m - s - x - 1.5d, \qquad c_1 = Y_1 + s + Rx.$$

- Same log utility and discount factor $\beta = 1$. No borrowing or shorting: $s, x \geq 0$, with $x = 0$ if $d = 0$ and $c_0 > 0$.
- Safe gross return is 1. Stock gross return R is 0.5 or 1.7, each with probability one half, independent of earnings.
- Y_1 follows U or D. In the realized-resource panel, $m = Y_0$ with zero inherited assets; Y_0 and Y_1 are independent draws from the same process.
- Financial saving is $s + x$: it excludes entry fees and subsequent returns. All amounts are in thousands of dollars.

Optimize with and without entry, then choose the higher value. These are teaching assumptions, not estimated parameters of the life-cycle model.

◀ Back ▶ Expectations ▶ Resources

At the same cash, how does duration affect saving?



In this illustration, lasting upside lowers saving even without current stock investment. At the same cash: $69 less in year 1 and $41 less in year 10. The one-year response is zero because future earnings are already baseline at the first new choice.

◀ Back ▶ Entry at the same cash ▶ All diagnostics

Which comparison answers which question?

How do later first-entry rates change? Annual first entry

Expectations: do choices change at the same state? Entry decisions Saving

Dynamic paths: saving can change later cash

Duration	Entry	Wealth	Budget
One year	Decomposition	Decomposition	Accounting
Five years	Decomposition	Decomposition	Accounting
Working life	Decomposition	Decomposition	Accounting

All comparisons use the same initial nonholders and fixed parameters. The shocks illustrate mechanisms. Earnings-risk coefficients are untargeted.

◀ Back ▶ Experiment design

Lasting upside: fewer first-time entrants in later years



Fewer people first enter from year 3 onward under lasting upside.
The early entry lead shrinks. Cumulative first entry still counts those who entered earlier.

◀ Back ▶ Cumulative entry ▶ Same-cash decisions

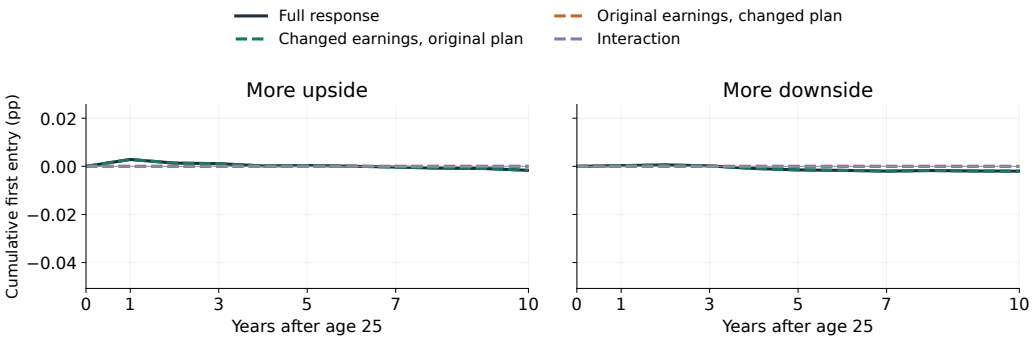
Illustration: lasting upside encourages entry at the same cash



The entry encouragement remains in year 10 at identical states.
Current nonholders on baseline paths, at identical resources and states. This includes re-entry. Probe choices do not alter subsequent states.

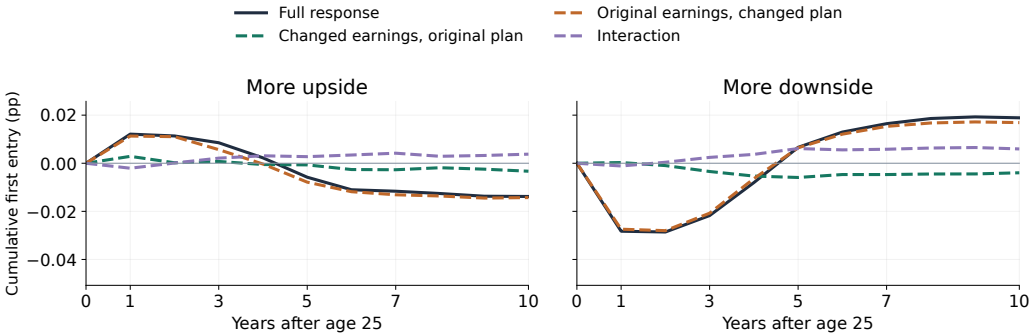
◀ Back ▶ Same-cash saving ▶ All diagnostics

One year: earnings paths and entry



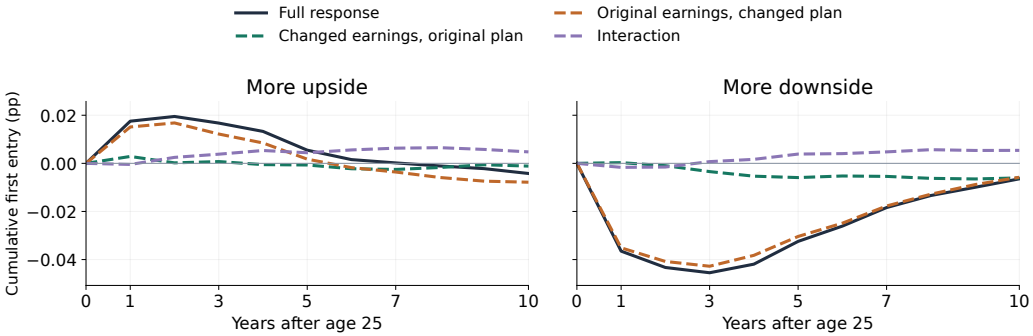
Only realized resources change at the first new decision.
The one-year change has ended for all future earnings. Rule-only and interaction responses are exactly zero. Tiny entry signs are grid-sensitive.

Five years: earnings paths and entry



The changed plan drives most of the later cumulative-entry reversal.
The crossed rule comparison includes saving and the resources it creates over time. It is not an equal-cash entry probe.

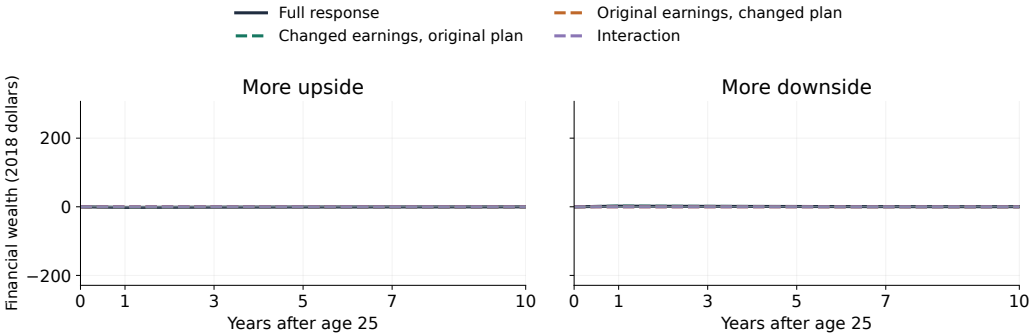
Working life: earnings paths and entry



The full cumulative-entry response ends close to zero.
Changed earnings and changed plans interact. The changed plan includes saving, investment and the later resources they produce.

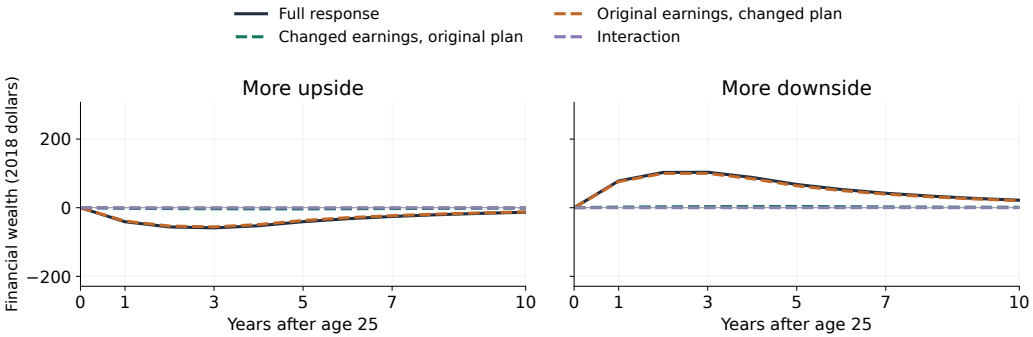
◀ Back ▶ Entry ▶ Wealth ▶ Budget

One year: earnings paths and wealth



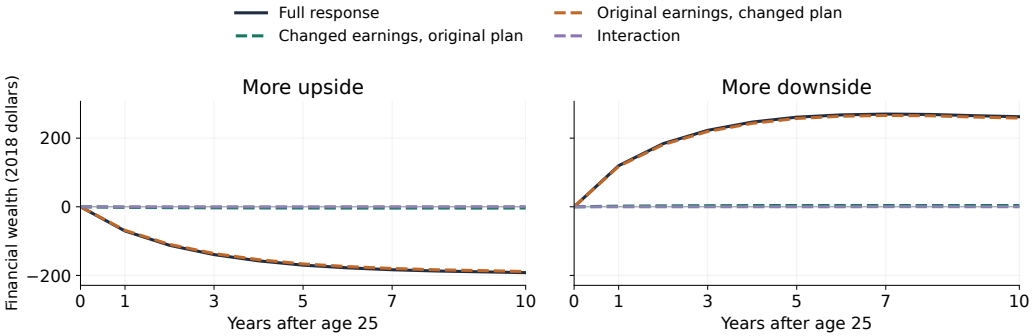
The small wealth effect comes through realized resources.
Future earnings prospects are baseline at the first new choice. Households use the original rule at their changed cash.

Five years: earnings paths and wealth



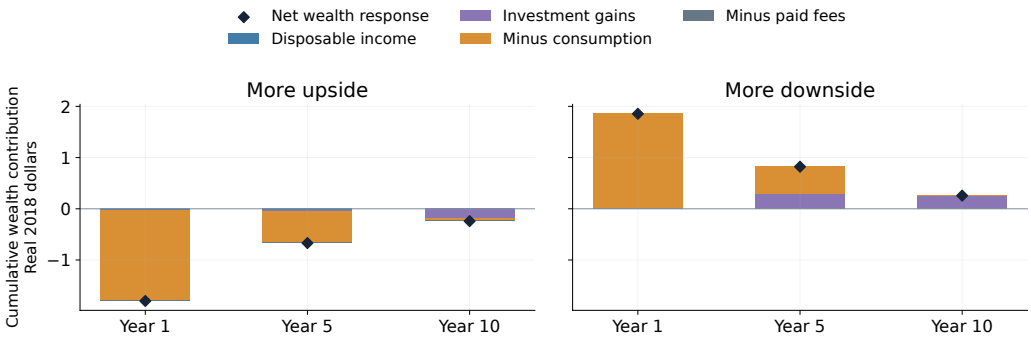
Earlier choices leave a wealth gap after the regime ends.
Rules coincide at the same state from year 5. The financial resources reached along each path can still differ.

Working life: earnings paths and wealth



The changed plan accounts for most of the persistent wealth gap.
At year 10 under upside, the plan-only contrast is about -$189. Changed earnings with the original plan contribute about -$4.

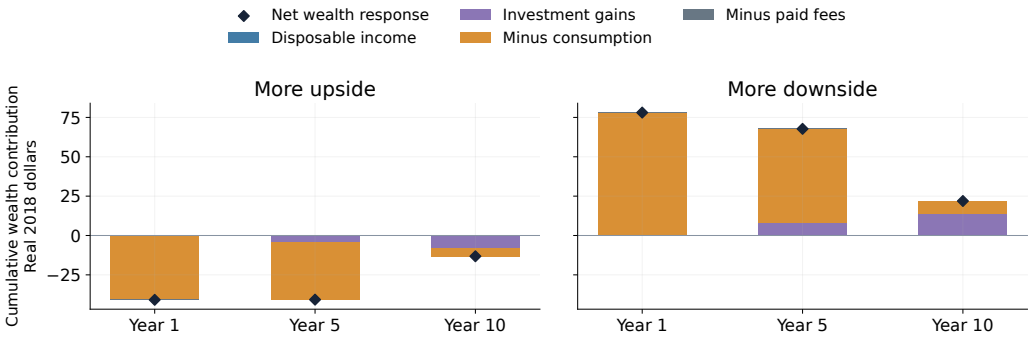
One year: where does the wealth difference come from?



With one year, the budget response is small.
After current earnings arrive, future prospects are already baseline. The current income draw changes resources, consumption and saving.

◀ Back ▶ Wealth decomposition ▶ Saving at the same cash

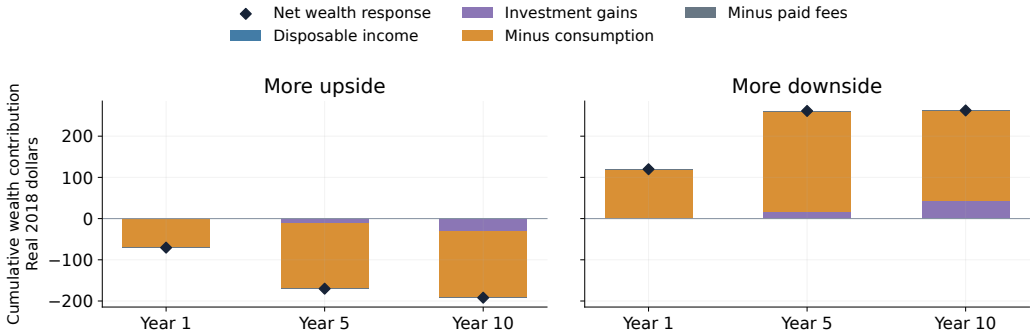
Five years: where does the wealth difference come from?



Consumption explains most of the initial wealth response.
As the temporary change ends, spending responses diminish. Earlier saving and investment choices still affect wealth.

◀ Back ▶ Wealth decomposition ▶ Saving at the same cash

Working life: where does the wealth difference come from?



Higher consumption accounts for most of the upside wealth decline.
At year 10: about -$163 from consumption, -$28 from investment gains and -$0.18 from fees.

◀ Back ▶ Wealth decomposition ▶ Saving at the same cash